GET it digital

Modul 9: Semiconductor devices

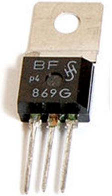
Stand: 18. Februar 2026

License

Learning objectives: Semiconductor

The students can

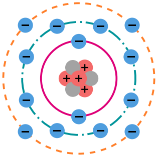
- explain the relationships between solids and the band model.
- describe processes within semiconductors.
- name different semiconductor materials and their properties.


BF
869G

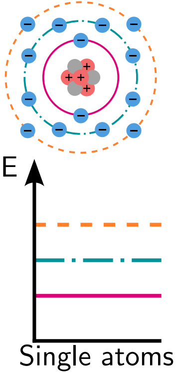
Einführung

- Semiconductor devices are central to the functioning of numerous electronic devices such as computers, mobile phones, and solar cells.
- They are based on materials that are neither good conductors nor good insulators. Their mode of operation can be explained using the band model.
- This chapter explains the fundamentals of semiconductor technology.

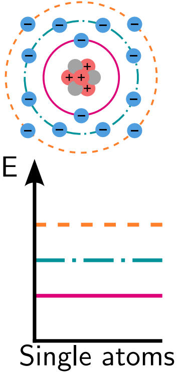
Band model



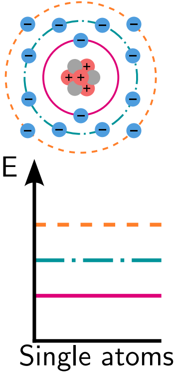
- The band model describes the electronic properties of solids and classifies the energy states of electrons into energy bands.



- The band model describes the electronic properties of solids and classifies the energy states of electrons into energy bands.
- It helps to understand and explain the electrical, optical and magnetic properties of materials.



- The band model describes the electronic properties of solids and classifies the energy states of electrons into energy bands.
- It helps to understand and explain the electrical, optical and magnetic properties of materials.
- The model enables the explanation of conduction, insulation, semiconductor behaviour, and surface and interface states.



- The band model describes the electronic properties of solids and classifies the energy states of electrons into energy bands.
- It helps to understand and explain the electrical, optical and magnetic properties of materials.
- The model enables the explanation of conduction, insulation, semiconductor behaviour, and surface and interface states.
- The illustration on the left shows the relationship between Bohr's atomic model and the band model.

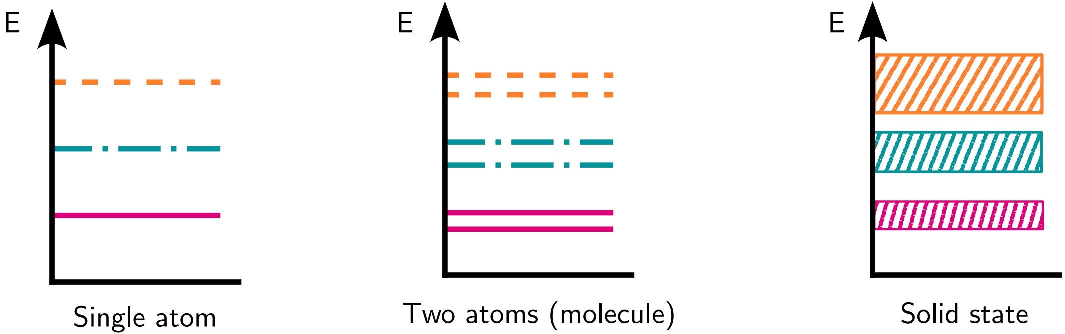


Figure: **Übergang von Energieniveaus zu Bändern** Relationship between possible energy states depending on the number of atoms. From left to right: energy states of electrons in a single atom, two atoms (molecule), and a solid.

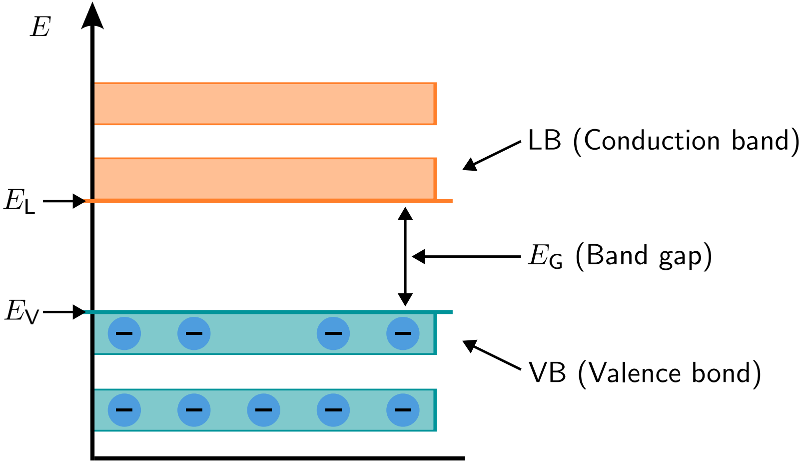


Figure: **Band model of a material at 0 K** Representation of the energy bands with corresponding labels and their alternative names.

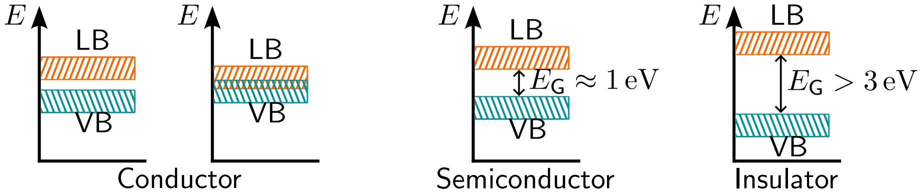


Figure: **Band model of various semiconductor materials** From left to right: Conductor without and with overlap, semiconductor, and insulator

Classification of materials

Key point:

- Charge carriers can only occupy defined energy levels in the solid state.

Key point:

- Charge carriers can only occupy defined energy levels in the solid state.
- At $T = 0\,\mathrm{K}$, the valence band is the highest occupied energy level; the conduction band above it contains no free charge carriers.

Key point:

- Charge carriers can only occupy defined energy levels in the solid state.
- At $T = 0\,\mathrm{K}$, the valence band is the highest occupied energy level; the conduction band above it contains no free charge carriers.
- Materials can be categorized via the band gap.

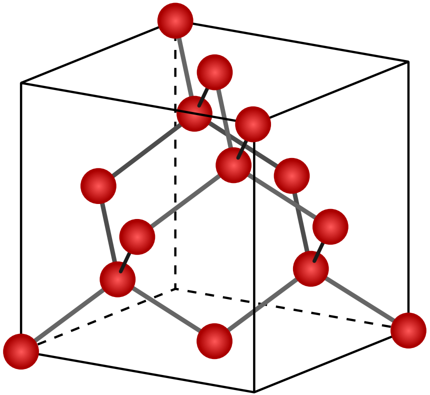


Figure: **Diamond lattice structure of silicon**

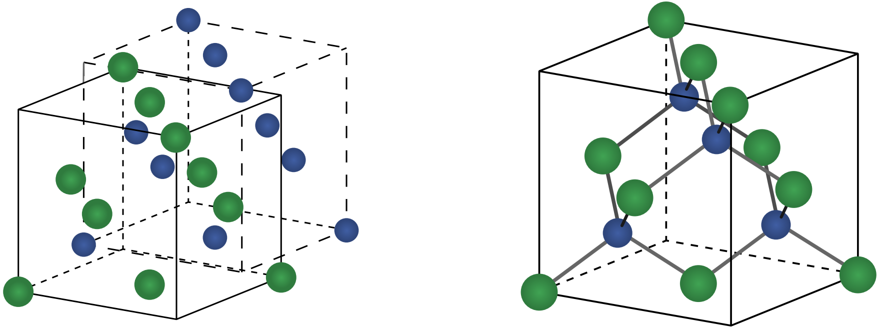


Figure: **Zinc blende lattice structure of GaAs**

Key point:

- Silicon is the most widely used semiconductor material.

Key point:

- Silicon is the most widely used semiconductor material.
- Compound semiconductors consist of two materials that exhibit semiconductor behavior in a specific combination.

Key point:

- Silicon is the most widely used semiconductor material.
- Compound semiconductors consist of two materials that exhibit semiconductor behavior in a specific combination.
- Silicon is from group IV of the periodic table, compound semiconductors are typically from groups III and V.

Semiconductor materials

Material	Band-gap (eV)	Charge-carrier density (cm^{-3})	Electron-mobility (cm^2/Vs)	Hole-mobility (cm^2/Vs)	Applications
Si	1,1	$10^{10} - 10^{15}$	1500	450	Microchips, Solar cells, sensors
Ge	0,7	$10^{13} - 20^{19}$	3900	1900	Transistors, Infrared detectors
GaAs	1,43	$10^{6} - 10^{8}$	8500	400	High-frequency circuits, LEDs, laser diodes

Table: Properties of various semiconductor materials

Semiconductor materials

Material	Band-gap (eV)	Charge-carrier density (cm^{-3})	Electron-mobility (cm^2/Vs)	Hole-mobility (cm^2/Vs)	Applications
InP	1,35	$10^{16} - 10^{17}$	5000 - 7000	200 - 400	Optoelectronics, solar cells
GaN	3,4	$10^{13} - 10^{17}$	1000 - 2000	50- 200	Power-electronics, LED lighting, displays
SiC	3,0	$10^{12} - 10^{16}$	800 - 1200	200 - 400	Power-electronics, high-temperature-applications

Table: Properties of various semiconductor materials

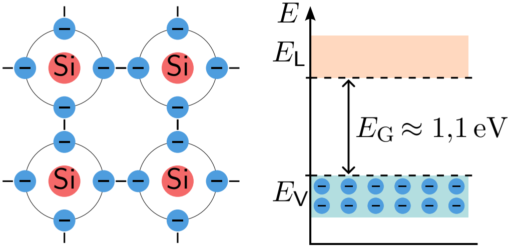


Figure: **Lattice structure and band model of Si** pure silicon at T = 0 K

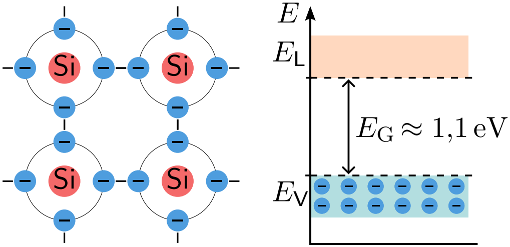


Figure: **Lattice structure and band model of Si** pure silicon at T = 0 K

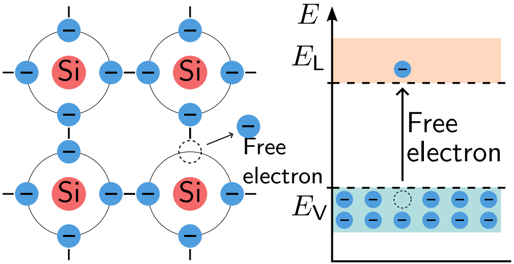


Figure: **Lattice structure and band model of Si** pure silicon at T ¿ 0 K

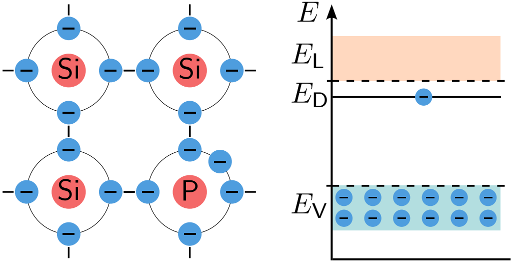


Figure: **Lattice structure and band model of n-doped Si** N-doped silicon at T = 0 K.

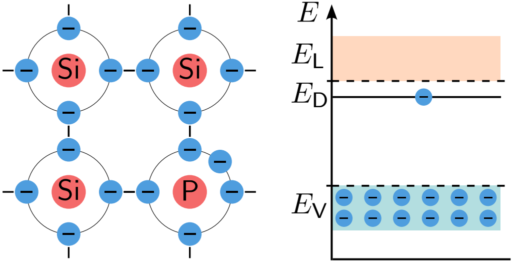


Figure: **Lattice structure and band model of n-doped Si** N-doped silicon at T $=$ 0 K.

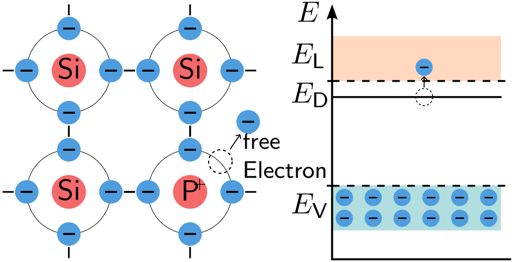


Figure: **Lattice structure and band model of n-doped Si** N-doped silicon at T $>$ 0 K.

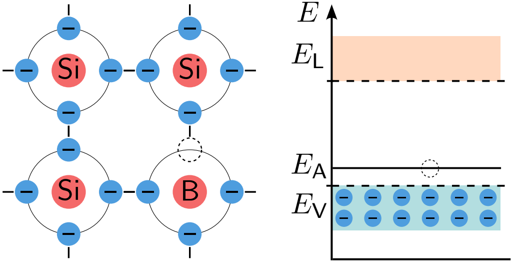


Figure: **Lattice structure and band model of p-doped Si** p-doped silicon at T = 0 K.

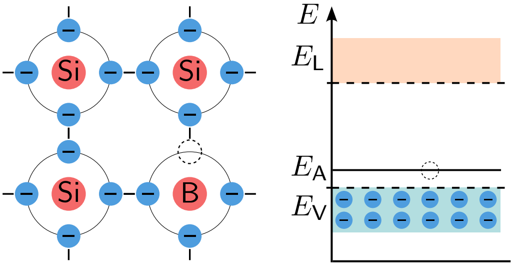


Figure: **Lattice structure and band model of p-doped Si** p-doped silicon at T = 0 K.

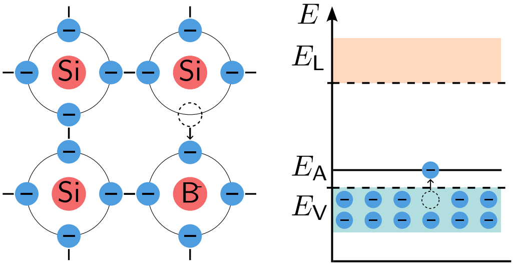


Figure: **Lattice structure and band model of p-doped Si** p-doped silicon at T > 0 K.

Key point:

- Thermal excitation can generate free electrons that contribute to charge carrier transport.

Key point:

- Thermal excitation can generate free electrons that contribute to charge carrier transport.
- Doping is the targeted introduction of foreign atoms with fewer valence electrons than the starting material.

Key point:

- Thermal excitation can generate free electrons that contribute to charge carrier transport.
- Doping is the targeted introduction of foreign atoms with fewer valence electrons than the starting material.
- Phosphorus can be used for n-doping and boron for p-doping.

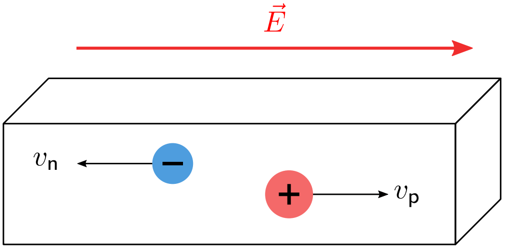


Figure: **Drift current within a semiconductor**

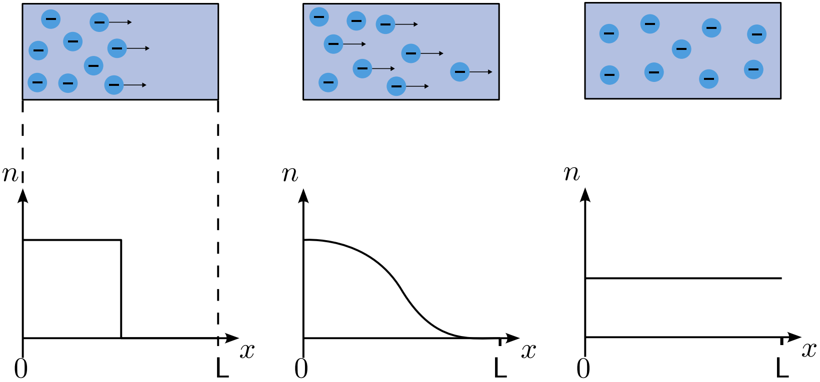


Figure: **Diffusion current within a semiconductor**

Charge carrier transport

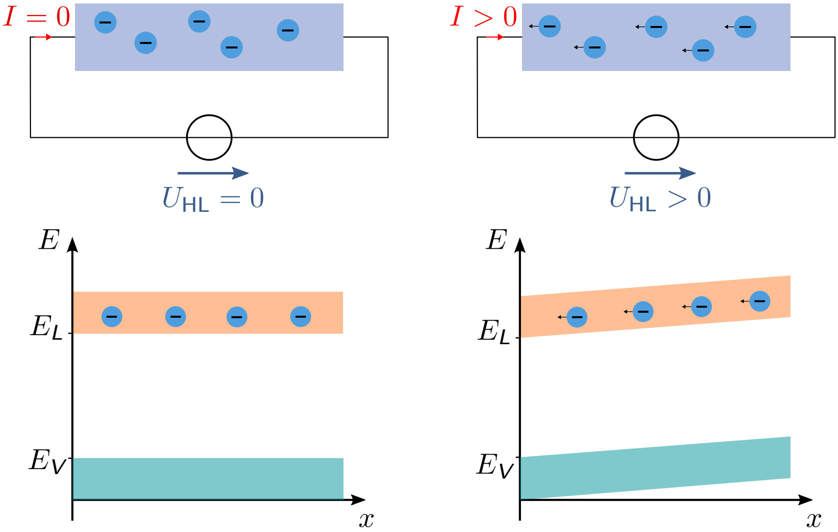


Figure: **Influence of voltages on semiconductors**

Key point:

- Drift current is the transport of charge carriers due to an electric field.

Charge carrier transport

Key point:

- Drift current is the transport of charge carriers due to an electric field.
- Diffusion current is the movement of charge carriers due to a difference in concentration.

Key point:

- Drift current is the transport of charge carriers due to an electric field.
- Diffusion current is the movement of charge carriers due to a difference in concentration.
- Drift current and diffusion current together make up the total current.

Key point:

- Drift current is the transport of charge carriers due to an electric field.
- Diffusion current is the movement of charge carriers due to a difference in concentration.
- Drift current and diffusion current together make up the total current.
- An external stress causes a shift in the band model.

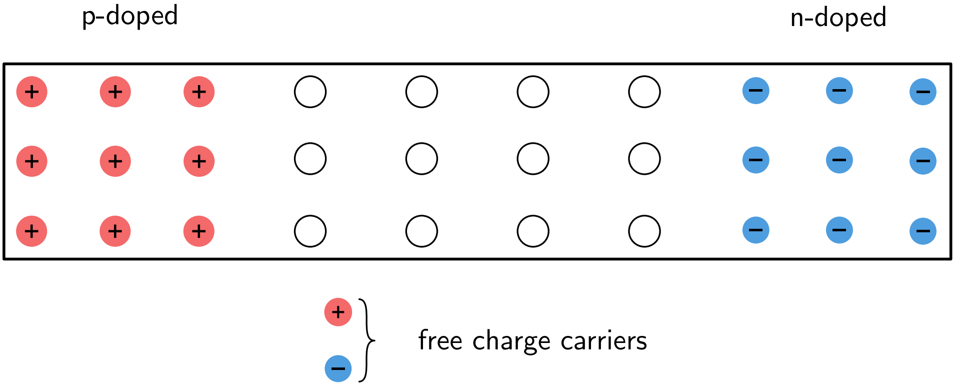


Figure: **Processes within a pn junction.**

pn-Junction

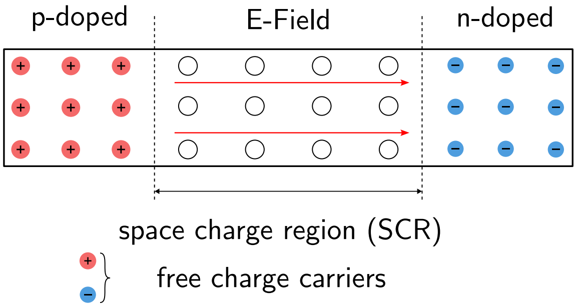


Figure: **Processes within a pn junction.**

- The binding of holes and electrons to their atoms limits the diffusion current.

pn-Junction

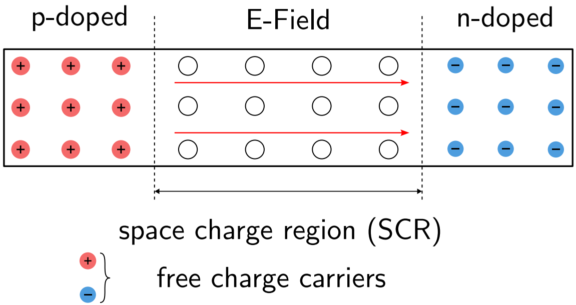


Figure: **Processes within a pn junction.**

- The binding of holes and electrons to their atoms limits the diffusion current.
- The area in the middle is called the space charge zone (SCZ). There are no free charge carriers there, as the electrons and holes neutralize each other.

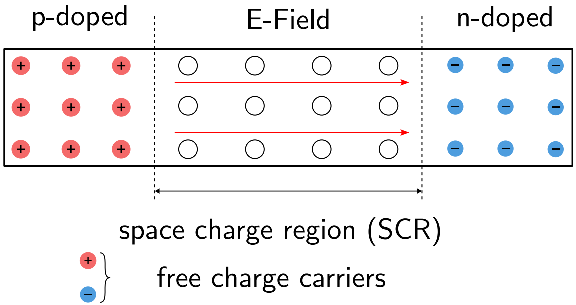


Figure: **Processes within a pn junction.**

- The binding of holes and electrons to their atoms limits the diffusion current.
- The area in the middle is called the space charge zone (SCZ). There are no free charge carriers there, as the electrons and holes neutralize each other.
- An electric field is created between the p- and n-doped areas.

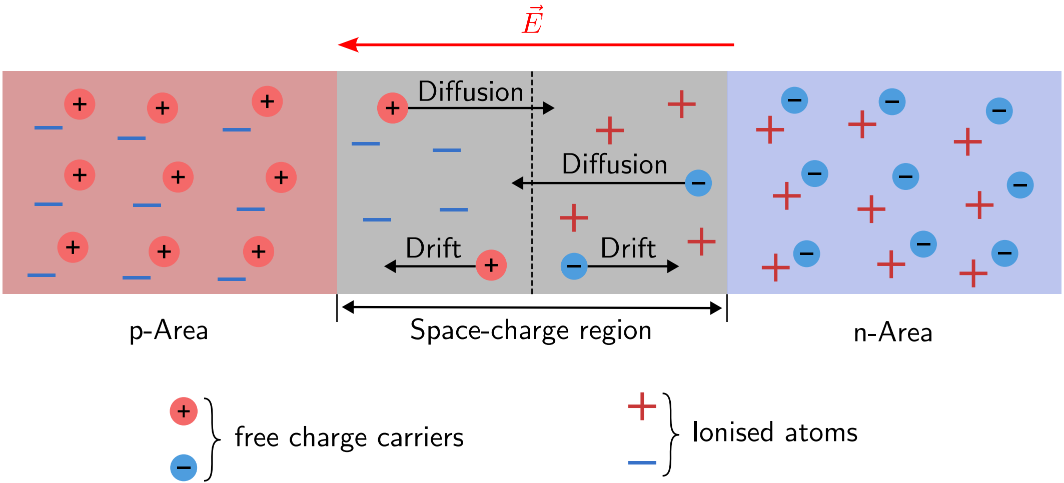


Figure: **Processes within the pn junction.**

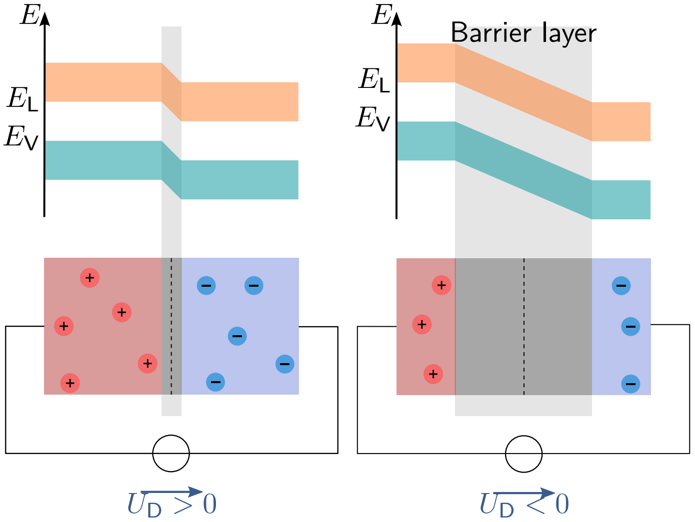


Figure: **Influence of voltages on pn junction.** Space charge region at different applied voltages. Left: voltage in the forward direction. Right: voltage against the forward direction, reverse direction.

pn-Junction

Key point:

- The pn junction combines a p- and n-doped semiconductor layer.

Key point:

- The pn junction combines a p- and n-doped semiconductor layer.
- There are no free charge carriers in the transition region; this region is called the space charge region.

Key point:

- The pn junction combines a p- and n-doped semiconductor layer.
- There are no free charge carriers in the transition region; this region is called the space charge region.
- In the flow direction, the RLZ is reduced, and in the reverse direction, it is increased.

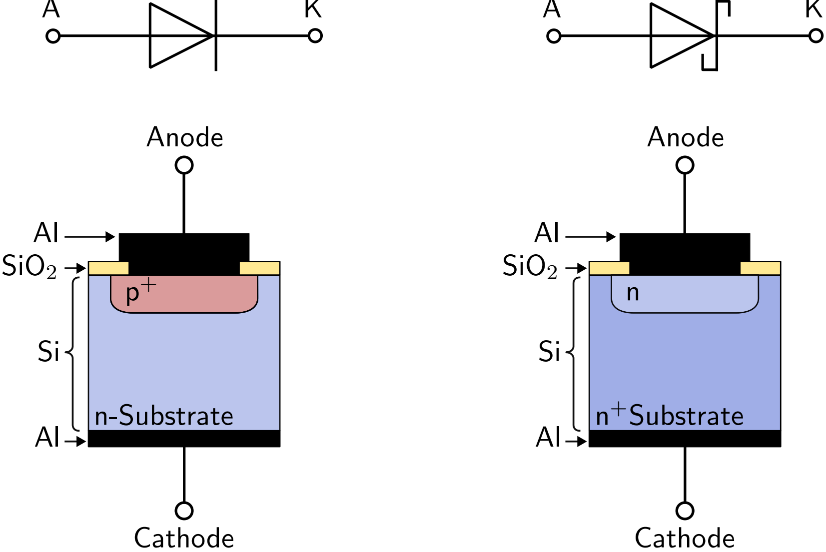


Figure: **Layer structure of diodes.** Structure and comparison of a classic silicon diode (left) and Schottky diode (right). Silicon dioxide ($\mathrm{SiO_2}$) is a natural oxide of silicon and is used for insulation. Aluminium (Al) is a typical metal for electrical contacts.

Key point: Semiconductor devices

The students can

- select components for various applications.
- describe the structure of various semiconductor devices.
- determine operating points and calculate missing component values.
- name applications for individual components.

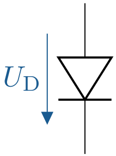


Figure: **Silizium Diode.** V.l.n.r. Schaltzeichen, Bauform und Querschnitt einer Silizium Diode.

Diode

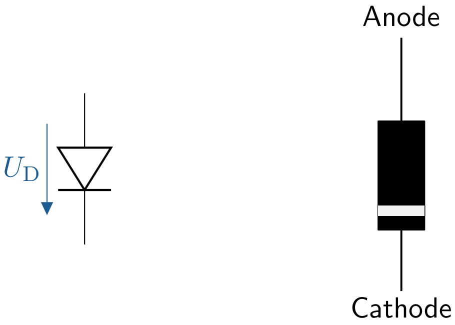


Figure: **Silizium Diode.** V.l.n.r. Schaltzeichen, Bauform und Querschnitt einer Silizium Diode.

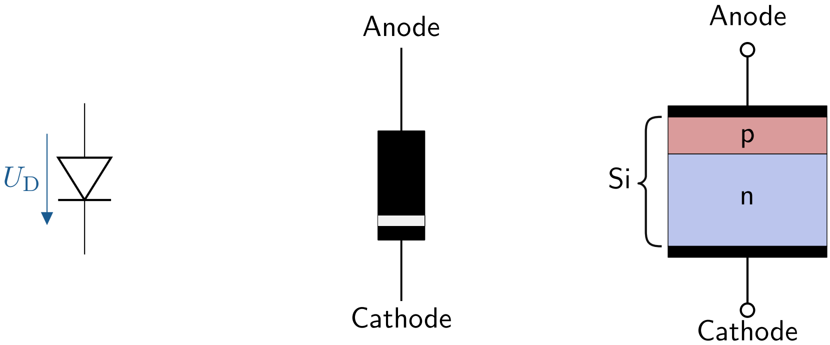
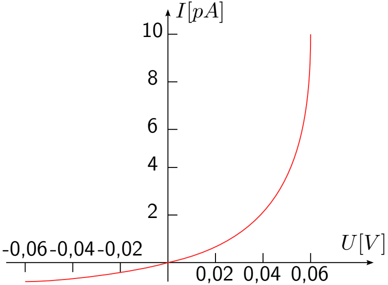


Figure: **Silizium Diode.** V.l.n.r. Schaltzeichen, Bauform und Querschnitt einer Silizium Diode.

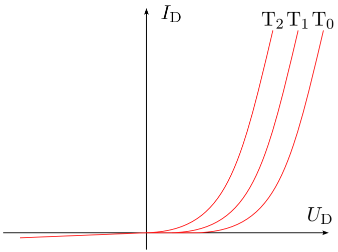
Diode characteristic curve



$$I_D = I_S \cdot (e^{\frac{U}{U_T}} - 1) = I_S e^{\frac{U_D}{U_T}} - I_S$$

- I_D: Diode current
- I_S: Blocking current
- U_D: Line voltage
- U_T: Temperature stress
- $U_T = \frac{k \cdot T}{e}$, at room temperature $U_T \approx 25\,\mathrm{mV}$

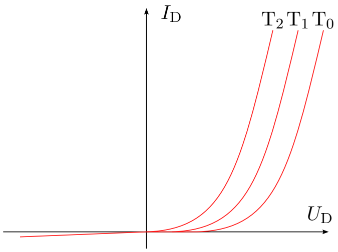
Temperature behavior of diodes



$I_\mathrm{D} = I_\mathrm{S} \cdot (e^{\frac{U}{U_\mathrm{T}}} - 1) = I_\mathrm{S} e^{\frac{U_\mathrm{D}}{U_\mathrm{T}}} - I_\mathrm{S}$ mit $U_\mathrm{T} = \frac{k \cdot T}{e}$

- Boltzmann constant $k = 1.380649 \cdot 10^{-23}\,\mathrm{J/K}$

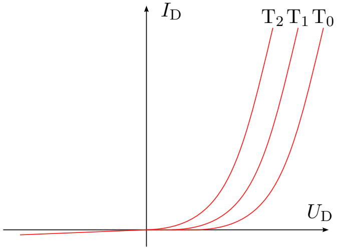
Temperature behavior of diodes



$I_\mathrm{D} = I_\mathrm{S} \cdot (e^{\frac{U}{U_\mathrm{T}}} - 1) = I_\mathrm{S} e^{\frac{U_\mathrm{D}}{U_\mathrm{T}}} - I_\mathrm{S}$ mit $U_\mathrm{T} = \frac{k \cdot T}{e}$

- Boltzmann constant $k = 1.380649 \cdot 10^{-23}\,\mathrm{J/K}$
- Every 5°C increase in temperature doubles the diode current.

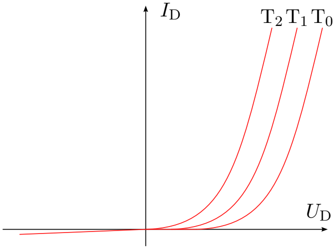
Temperature behavior of diodes



$$I_D = I_S \cdot (e^{\frac{U}{U_T}} - 1) = I_S e^{\frac{U_D}{U_T}} - I_S \text{ mit } U_T = \frac{k \cdot T}{e}$$

- Boltzmann constant $k = 1.380649 \cdot 10^{-23}\,\mathrm{J/K}$
- Every 5°C increase in temperature doubles the diode current.
- The forward voltage U_D decreases by approx. 2 mV per °C at constant diode current.

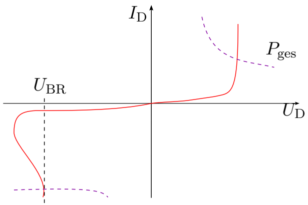
Temperature behavior of diodes



$I_D = I_S \cdot (e^{\frac{U}{U_T}} - 1) = I_S e^{\frac{U_D}{U_T}} - I_S$ mit $U_T = \frac{k \cdot T}{e}$

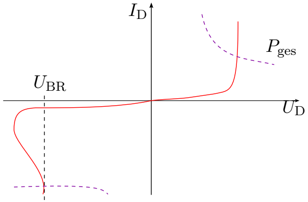
- Boltzmann constant $k = 1.380649 \cdot 10^{-23}\,\mathrm{J/K}$
- Every 5°C increase in temperature doubles the diode current.
- The forward voltage U_D decreases by approx. 2 mV per °C at constant diode current.
- $T_2 > T_1 > T_0$

Load limits of a diode



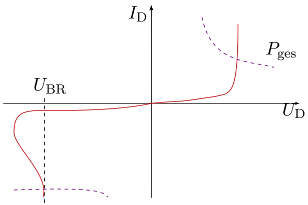
- A diode can withstand more voltage in the reverse direction than in the forward direction.

Load limits of a diode



- A diode can withstand more voltage in the reverse direction than in the forward direction.
- High voltages in the reverse direction can cause a breakdown.

Load limits of a diode



- A diode can withstand more voltage in the reverse direction than in the forward direction.
- High voltages in the reverse direction can cause a breakdown.
- The diode in reverse direction is irreversibly destroyed in the area marked in purple.

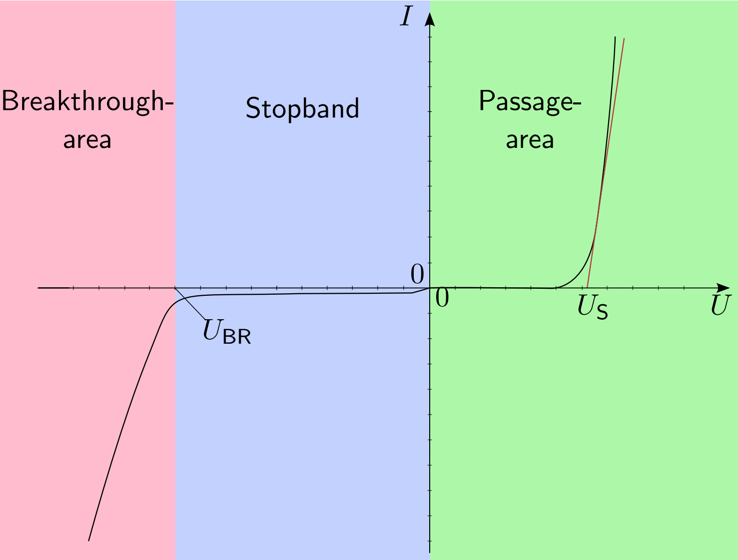


Figure: **Diode characteristic curve.** Relevant areas from left to right: breakthrough area, barrier area, and passage area

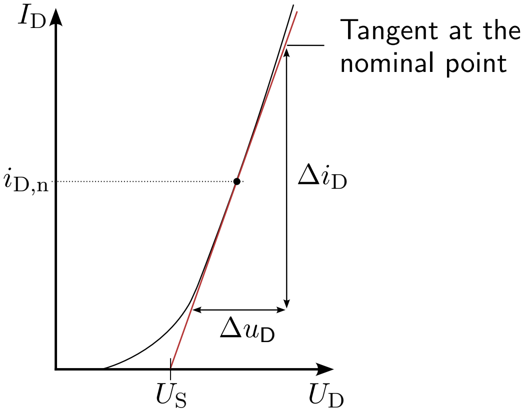
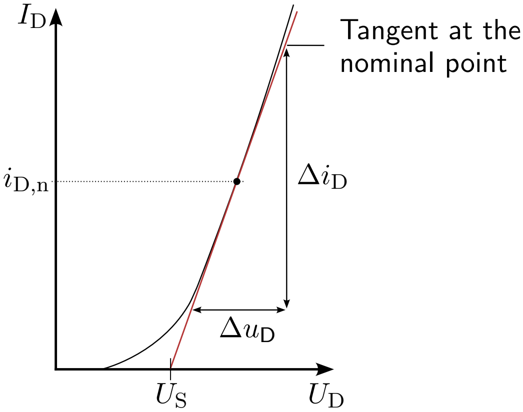


Figure: **Diode characteristic curve and equivalent circuit diagram.** Left: Diode characteristic curve with approximation for the differential resistance. Right: Diode equivalent circuit diagram with the ideal diode, voltage source, and differential resistance.

Diode – Electrical behavior



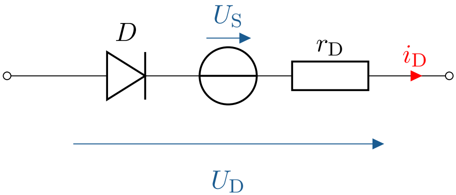
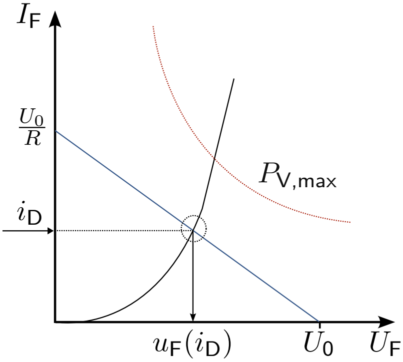


Figure: **Diode characteristic curve and equivalent circuit diagram.** Left: Diode characteristic curve with approximation for the differential resistance. Right: Diode equivalent circuit diagram with the ideal diode, voltage source, and differential resistance.



I_F
$\frac{U_0}{R}$
i_D
$P_{V,max}$
$u_F(i_D)$
U_0
U_F

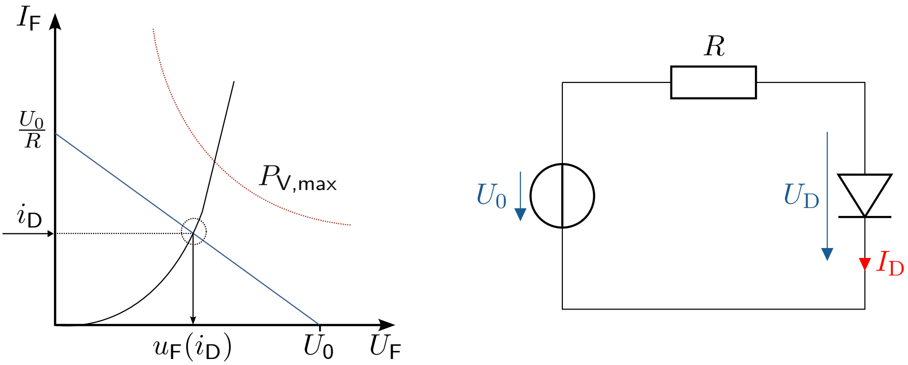


Figure: **Graphical determination of the operating point.** Left: Diode characteristic curve (black), resistance line (blue), and asymptote of maximum power dissipation (red) of the diode. Right: Circuit of the diode with series resistor.

Key point:

- The characteristic curve of the diode is highly nonlinear.

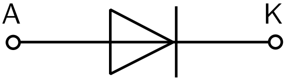
Key point:

- The characteristic curve of the diode is highly nonlinear.
- The behavior of the characteristic curve can be described in three areas: the breakthrough, blocking, and pass-through areas.

Key point:

- The characteristic curve of the diode is highly nonlinear.
- The behavior of the characteristic curve can be described in three areas: the breakthrough, blocking, and pass-through areas.
- Using an ideal diode, voltage source, and resistor, the real diode can be approximated in various areas.

Diode – Structure



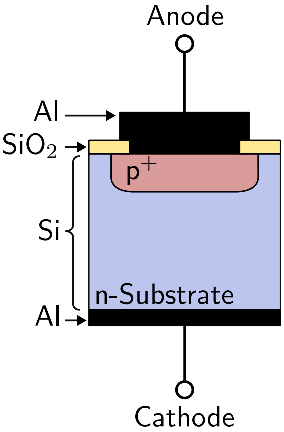


Figure: **Layer structure of diodes.** Structure and comparison of a classic silicon diode (left) and Schottky diode (right).

Diode – Structure

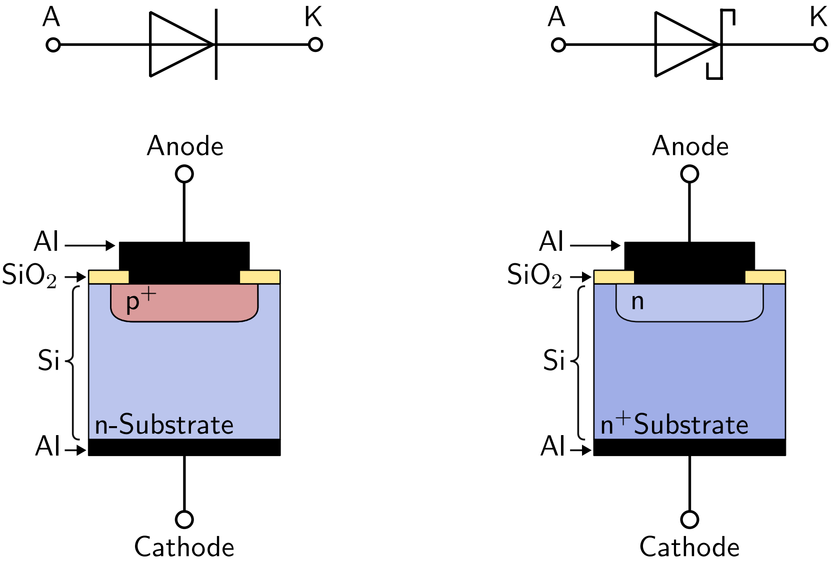
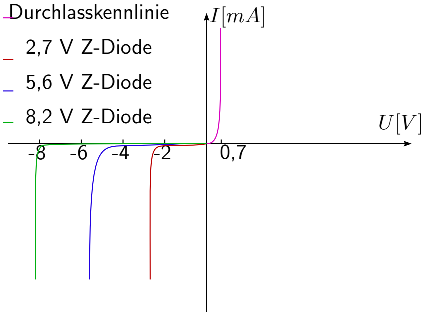


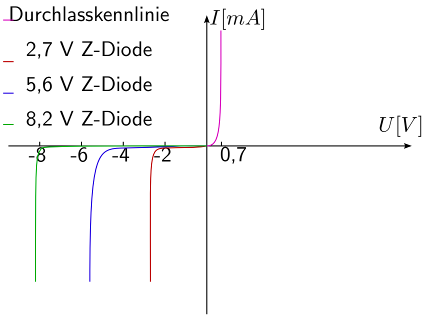
Figure: **Layer structure of diodes.** Structure and comparison of a classic silicon diode (left) and Schottky diode (right).

Characteristic curves of a Zener diode



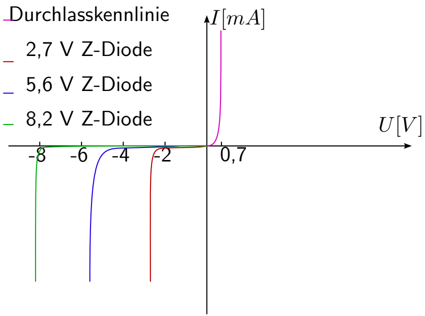
- Zener effect at a breakdown voltage of $< 5\,V$ (electrons tunnel in the reverse direction)

Characteristic curves of a Zener diode



- Zener effect at a breakdown voltage of $< 5\,V$ (electrons tunnel in the reverse direction)
- Avalanche effect at a breakdown voltage of $> 5\,V$ (avalanche breakdown effect)

Characteristic curves of a Zener diode



- Zener effect at a breakdown voltage of $< 5\,V$ (electrons tunnel in the reverse direction)
- Avalanche effect at a breakdown voltage of $> 5\,V$ (avalanche breakdown effect)
- U_{BR} possible in the range from $1.8\,V$ to $300\,V$.

Diode – Structure

Key point:

- Schottky diodes do not have a pn junction, but rather a Schottky contact.

Key point:

- Schottky diodes do not have a pn junction, but rather a Schottky contact.
- With a suitable combination of materials, an RLZ forms between the metal and the semiconductor.

Diode – special diodes

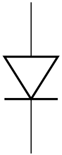
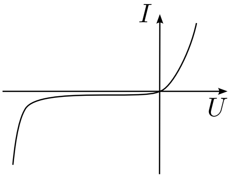
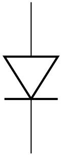
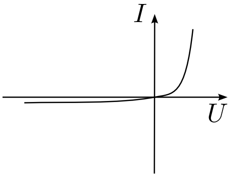
Designation	Symbol	Char. curve	Properties	Application
Rectifier-diode			High forward current	high reverse voltage, rectification
Switching-diode			Low breakdown resistance	high reverse resistance, short switching times

Table: **Overview of typical diodes.**

Diode – special diodes

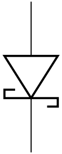
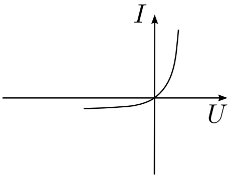
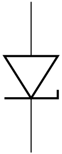
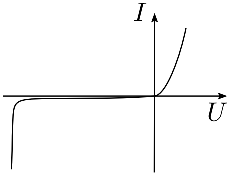
Designation	Symbol	Char. curve	Properties	Application
Schottky-diode			Low forward voltage	low reverse voltage, high-frequency rectifier, free-wheeling diode, switching power supplies
Z-Diode			defined breakdown voltage	stabilization of voltages, limitation

Table: **Overview of typical diodes.**

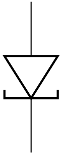
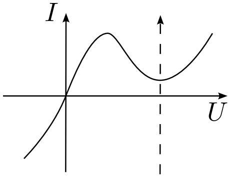
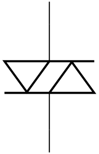
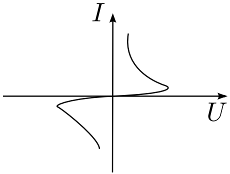
Designation	**Symbol**	**Char. curve**	**Properties**	**Application**
Tunneldiode			Negative differential resistance	de-attenuation of oscillating circuits, RF oscillator
Diac			Controlled breakdown	de-buffering, trigger diode

Table: **Overview of typical diodes.**

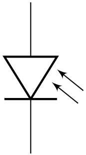
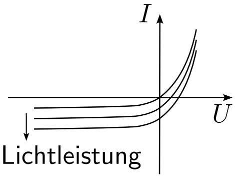
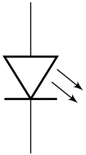
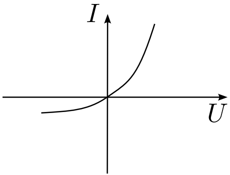
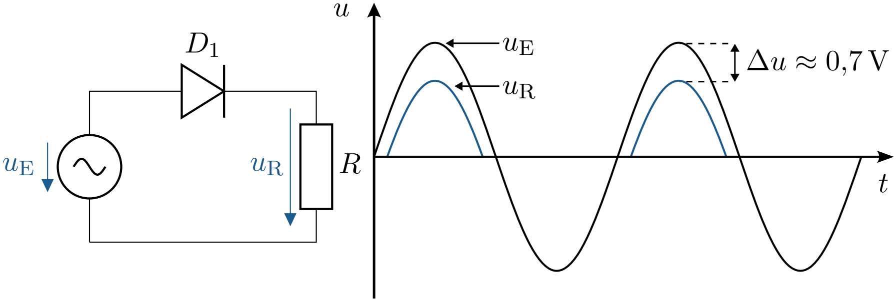
Designation	Symbol	Char. curve	Properties	Application
Photodiode			Current changes proportionally to light output	Photo receiver, measurement technology, solar cells
LED, Laserdiode			Passage current generates optical radiation	lighting, radiation source

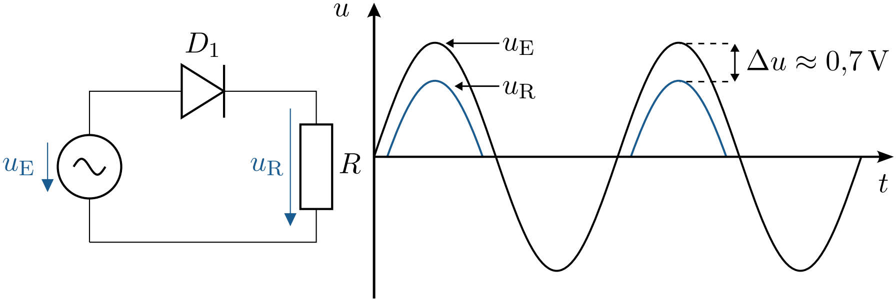
Table: **Overview of typical diodes.**

Diode for alternating voltage

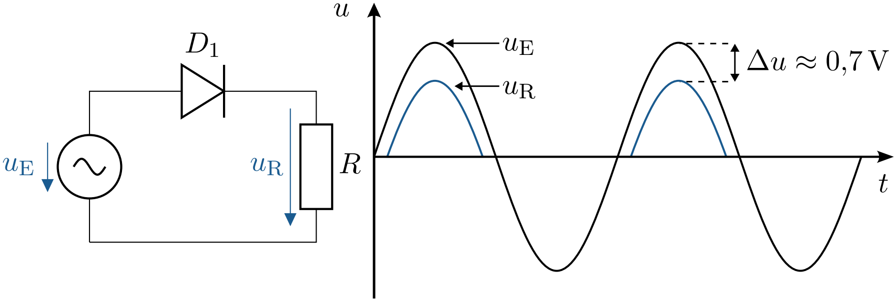


- In the reverse direction, the entire voltage drops across the diode.

Diode for alternating voltage

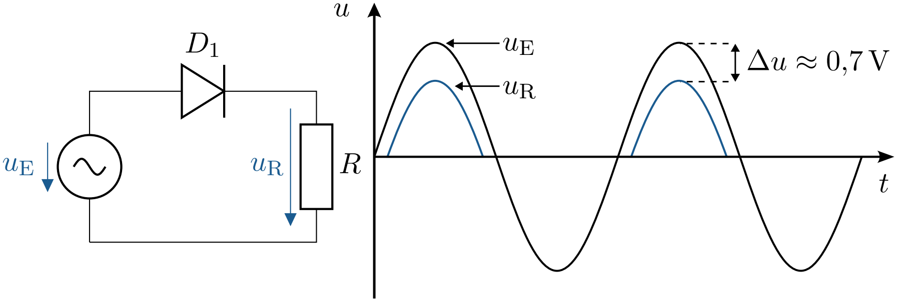


- In the reverse direction, the entire voltage drops across the diode.
- The positive half-wave of the alternating voltage is passed through. The forward voltage of the diode is subtracted.



- In the reverse direction, the entire voltage drops across the diode.
- The positive half-wave of the alternating voltage is passed through. The forward voltage of the diode is subtracted.
- The negative half-wave of the alternating voltage is blocked.

Diode for alternating voltage



- In the reverse direction, the entire voltage drops across the diode.
- The positive half-wave of the alternating voltage is passed through. The forward voltage of the diode is subtracted.
- The negative half-wave of the alternating voltage is blocked.
- The diode acts as a rectifier.

Single-phase rectifier

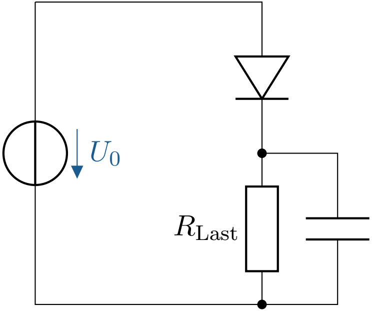


Figure: **Circuit of a single-phase rectifier.**

- A DC voltage cannot be generated with a diode alone.

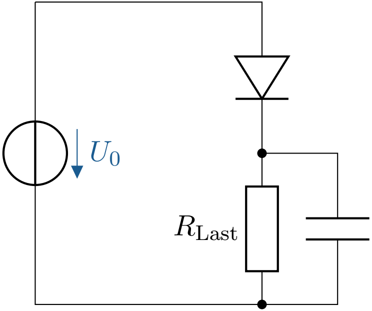


Figure: **Circuit of a single-phase rectifier.**

- A DC voltage cannot be generated with a diode alone.
- Signal must be smoothed by a capacitor.

Voltage curve of a single-phase rectifier

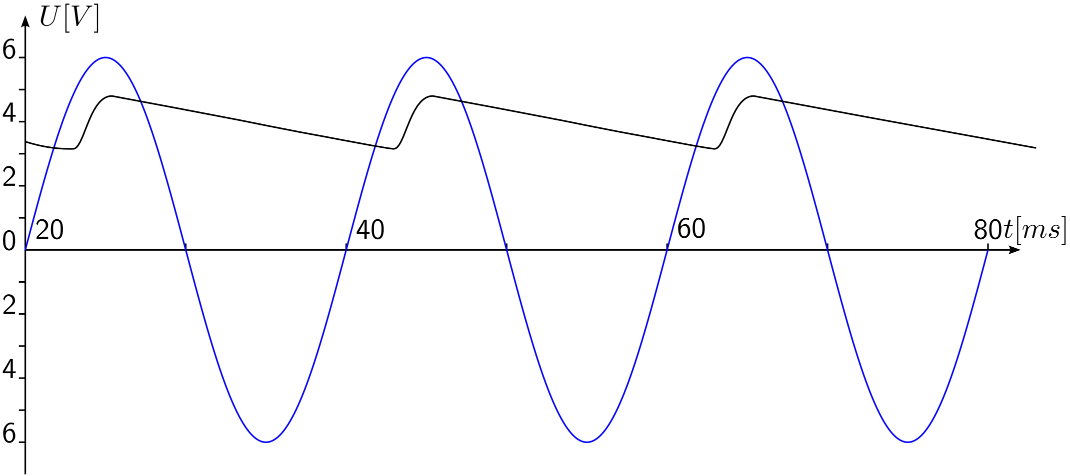


Figure: **Single-phase rectifier.** Voltage curve of a rectifier.

Current flow of a single-phase rectifier

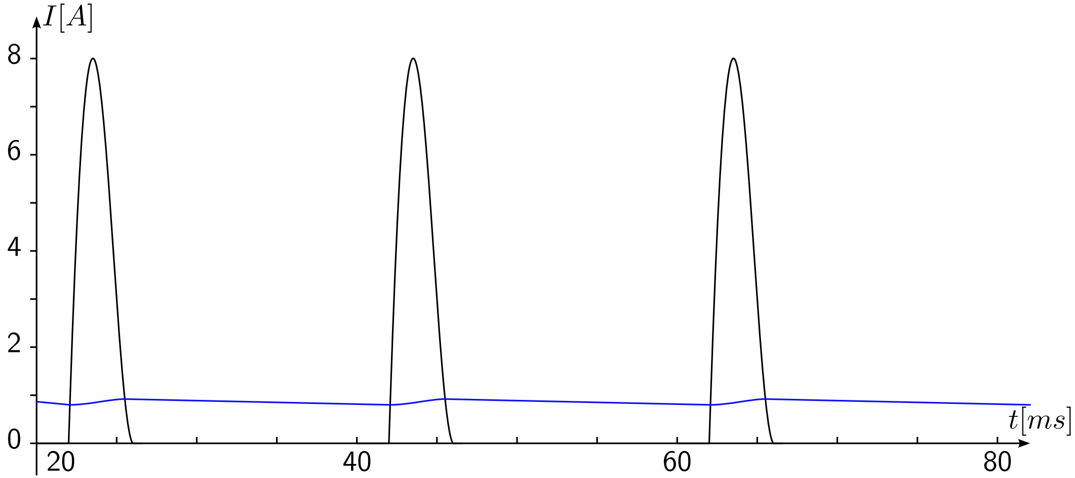


Figure: **Single-phase rectifier.** Current flow of a rectifier.

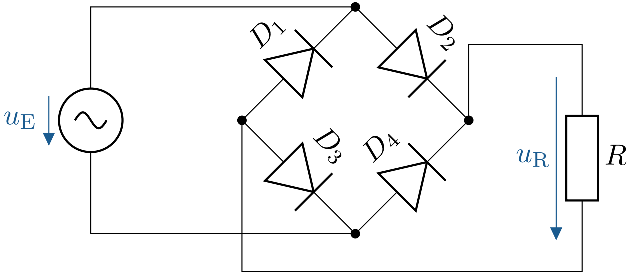


Figure: **Bridge rectifier circuit** Circuit of a bridge rectifier with applied load R.

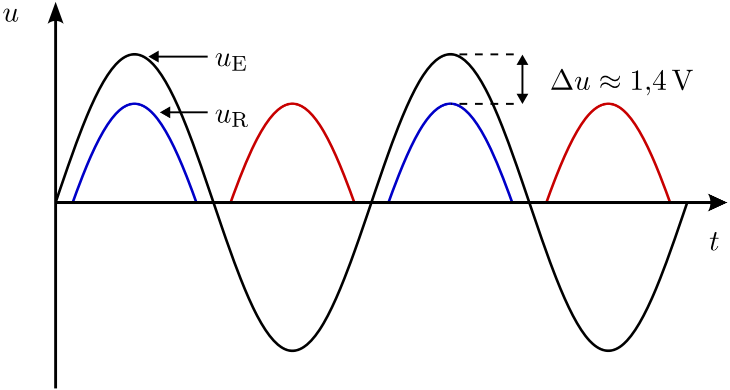


Figure: **Bridge rectifier signal waveforms.** Input voltage and load voltage curve of the bridge rectifier

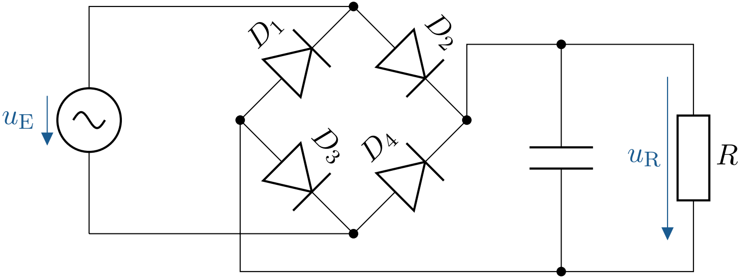


Figure: **Bridge rectifier circuit with smoothing capacitor**

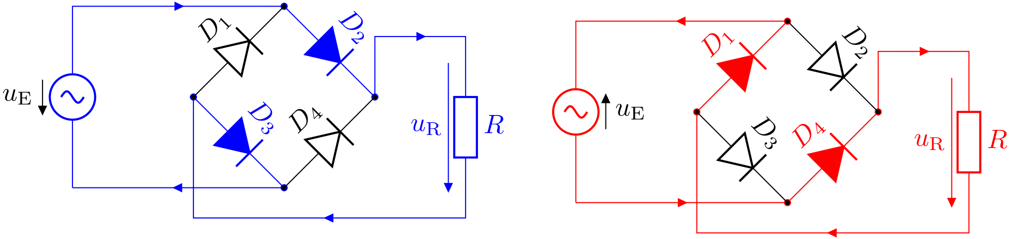


Figure: **Current paths in the bridge rectifier.** Left: Current path during the positive half-wave of the input voltage. Right: Current path during the negative half-wave of the input voltage.

Delon-circuit

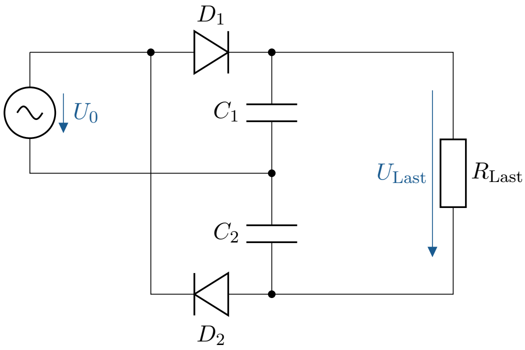


Figure: **Delon-circuit**

- Includes a rectifier circuit consisting of two diodes and two capacitors.

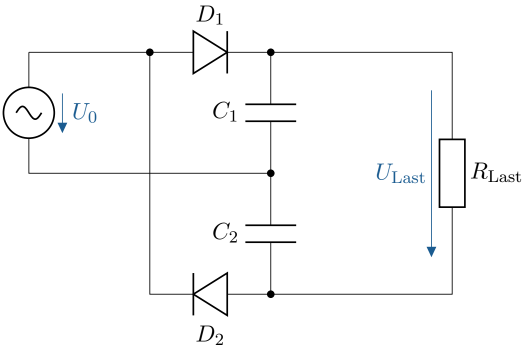


Figure: **Delon-circuit**

- Includes a rectifier circuit consisting of two diodes and two capacitors.
- The positive half-wave charges capacitor C_1 via diode D_1.

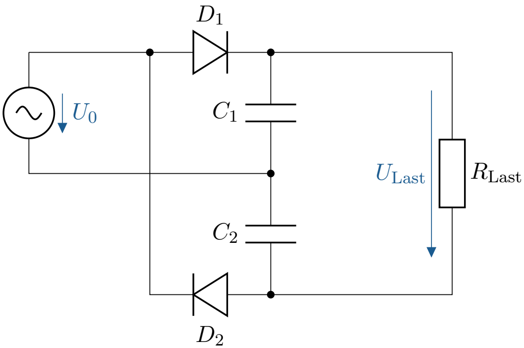


Figure: **Delon-circuit**

- Includes a rectifier circuit consisting of two diodes and two capacitors.
- The positive half-wave charges capacitor C_1 via diode D_1.
- The negative half-wave charges capacitor C_2 via diode D_2.

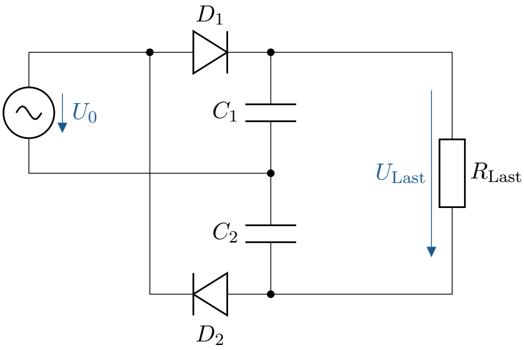


Figure: **Delon-circuit**

- Includes a rectifier circuit consisting of two diodes and two capacitors.
- The positive half-wave charges capacitor C_1 via diode D_1.
- The negative half-wave charges capacitor C_2 via diode D_2.
- Because the output is parallel to the capacitors connected in series, the voltage is doubled.

Key point:

- Diodes are used in rectifiers to convert alternating current into direct current.

Key point:

- Diodes are used in rectifiers to convert alternating current into direct current.
- In a half-wave rectifier, only one polarity of the half-waves is passed through.

Key point:

- Diodes are used in rectifiers to convert alternating current into direct current.
- In a half-wave rectifier, only one polarity of the half-waves is passed through.
- Bridge rectifiers use both polarities of the input voltage.

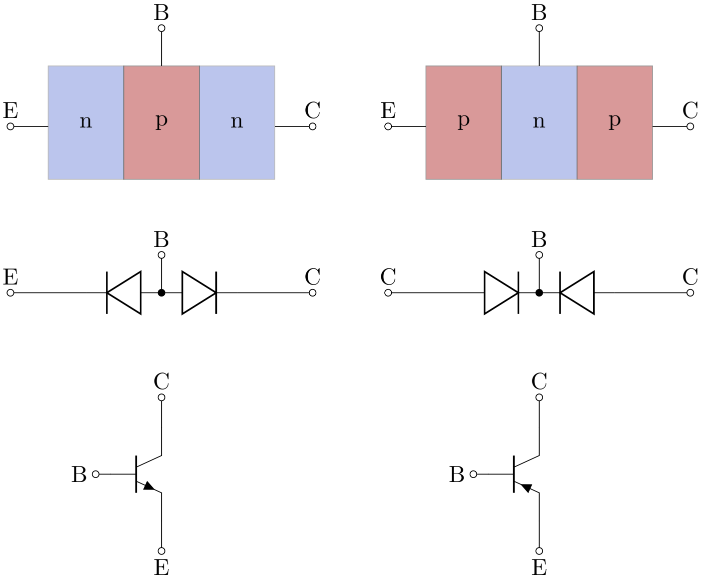


Figure: **Overview of bipolar transistors.** Simplified representation of the cross-section, logic,

Simplified representation of the cross-section, logic, and circuit symbols of npn and pnp transistors.

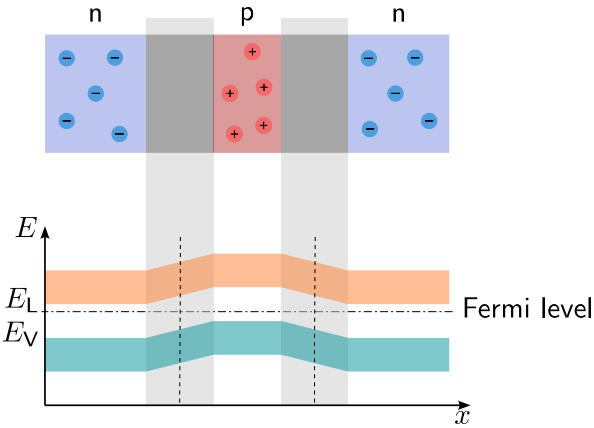


Figure: **Behavior of an unconnected bipolar transistor.** Cross-section of an unconnected NPN transistor and associated band model

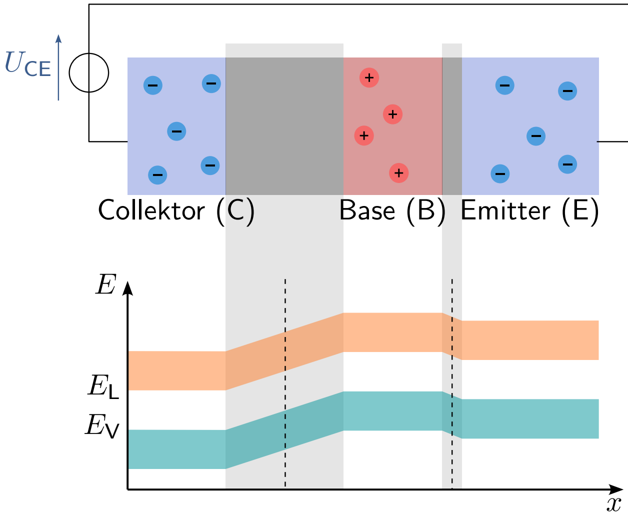


Figure: **Bipolar transistor with collector-emitter voltage.** Cross-section of an NPN transistor and associated band model with a positive voltage between the collector and emitter.

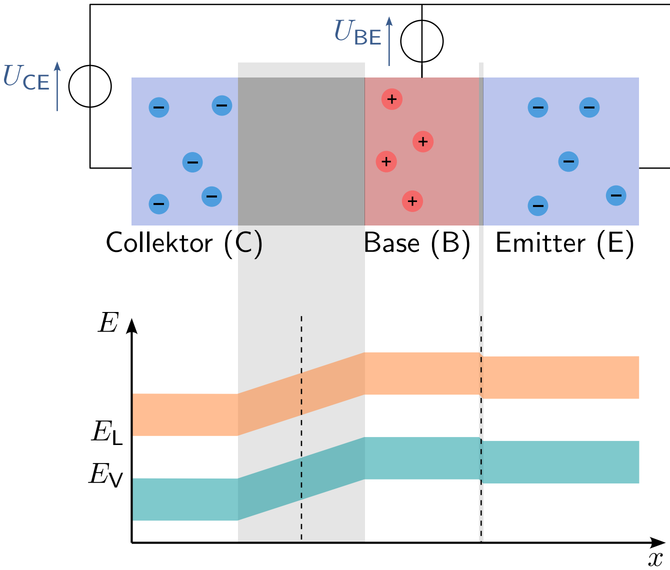


Figure: **Connected bipolar transistor** Cross-section of an NPN transistor and associated band model with a positive voltage between the collector and emitter as well as between the base

Key point:

- Bipolar transistors have two pn junctions.

Key point:

- Bipolar transistors have two pn junctions.
- There are three terminals: the collector (C), the base (B), and the emitter (E).

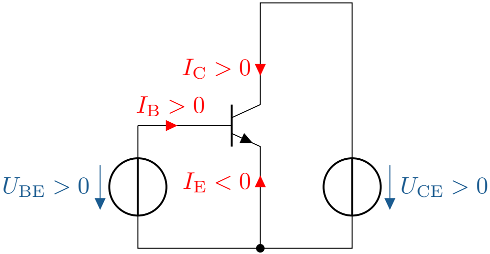


Figure: **Currents and voltages in bipolar transistors.** Relevant currents and voltages for npn and pnp transistors.

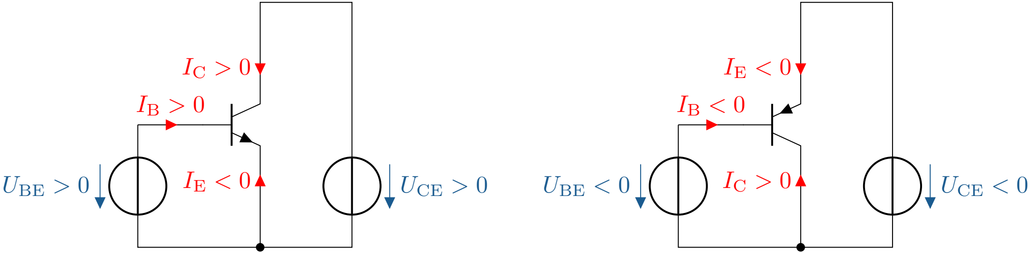


Figure: **Currents and voltages in bipolar transistors.** Relevant currents and voltages for npn and pnp transistors.

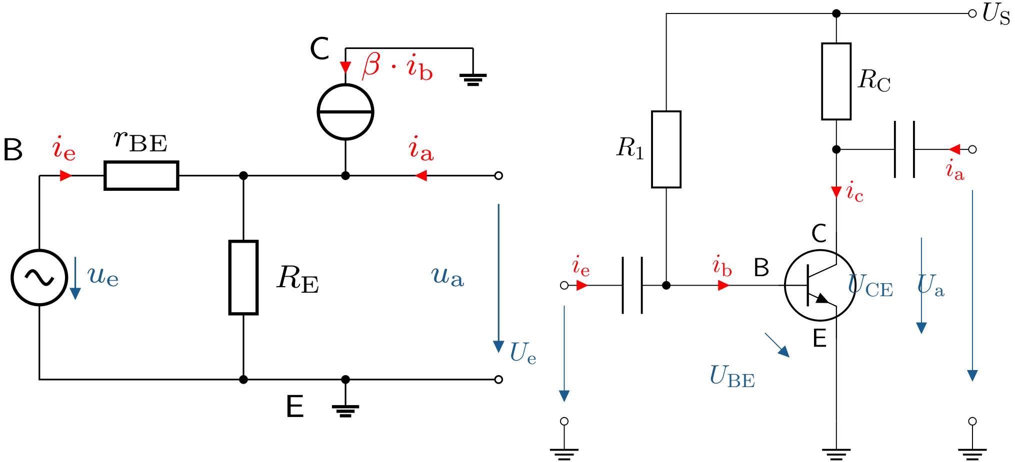


Figure: Basic circuit and equivalent circuit diagram of a collector circuit.

- Input resistance $r_E = \frac{u_e}{i_e} = \frac{R_1 r_{BE}}{R_1 + r_{BE}} = R_1 \parallel r_{BE}$

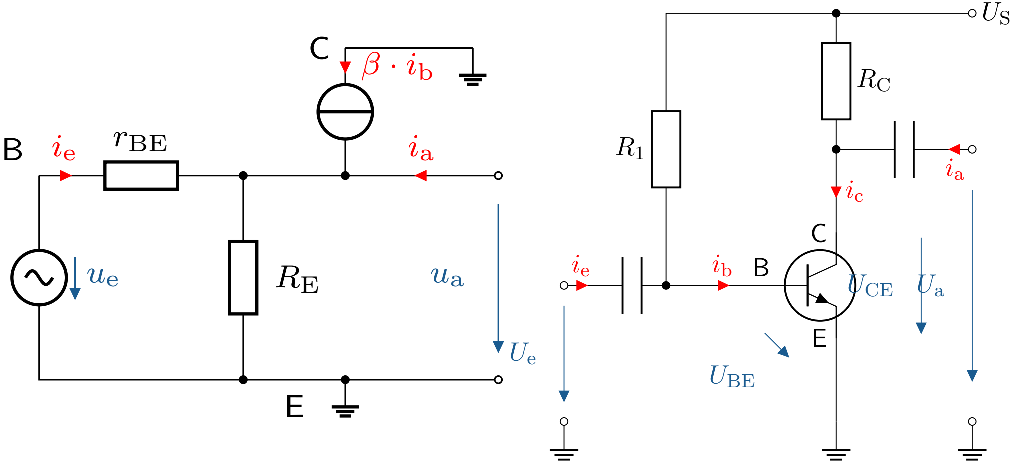
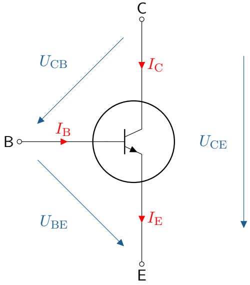


Figure: Basic circuit and equivalent circuit diagram of a collector circuit.

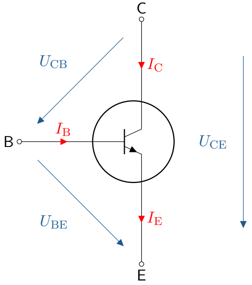
- Input resistance $r_E = \frac{u_e}{i_e} = \frac{R_1 r_{BE}}{R_1 + r_{BE}} = R_1 \parallel r_{BE}$
- Output resistance $r_A = \frac{u_e}{i_e} = \frac{R_C r_{CE}}{R_C + r_{CE}} = R_C \parallel r_{CE}$

Transistor voltages and currents



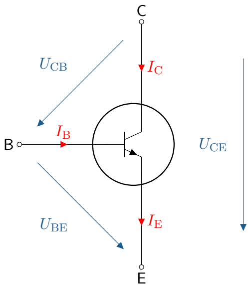
- Values in the switched-on state:

Transistor voltages and currents



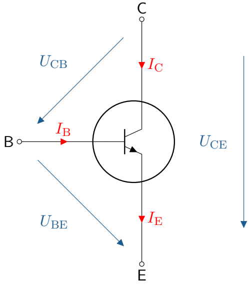
- Values in the switched-on state:
- $U_{BE} \approx 0.6\,\mathrm{V}$ (Base-emitter voltage)

Transistor voltages and currents



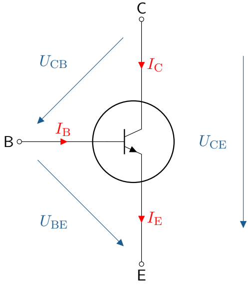
- Values in the switched-on state:
- $U_{BE} \approx 0.6\,\mathrm{V}$ (Base-emitter voltage)
- $U_{CB} \approx 2 ... 300\,\mathrm{V}$ (Collector base voltage)

Transistor voltages and currents



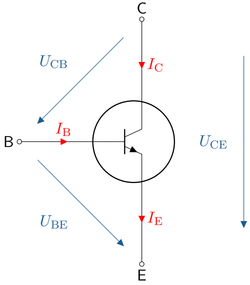
- Values in the switched-on state:
- $U_{BE} \approx 0.6\,\mathrm{V}$ (Base-emitter voltage)
- $U_{CB} \approx 2...300\,\mathrm{V}$ (Collector base voltage)
- $I_B = 1\,\%; I_C = 99\,\%; I_E = 100\,\%$

Transistor voltages and currents



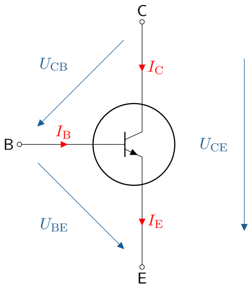
- Values in the switched-on state:
- $U_{\mathrm{BE}} \approx 0.6\,\mathrm{V}$ (Base-emitter voltage)
- $U_{\mathrm{CB}} \approx 2...300\,\mathrm{V}$ (Collector base voltage)
- $I_{\mathrm{B}} = 1\,\%; I_{\mathrm{C}} = 99\,\%; I_{\mathrm{E}} = 100\,\%$
- The node and mesh rules apply in the transistor:

Transistor voltages and currents

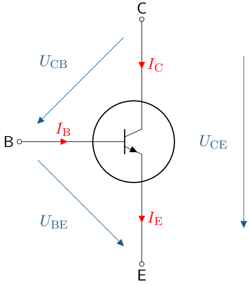


- Values in the switched-on state:
- $U_{\mathrm{BE}} \approx 0.6\,\mathrm{V}$ (Base-emitter voltage)
- $U_{\mathrm{CB}} \approx 2...300\,\mathrm{V}$ (Collector base voltage)
- $I_{\mathrm{B}} = 1\,\%; I_{\mathrm{C}} = 99\,\%; I_{\mathrm{E}} = 100\,\%$
- The node and mesh rules apply in the transistor:
- $U_{\mathrm{CE}} = U_{\mathrm{CB}} + U_{\mathrm{BE}} = 0$

Transistor voltages and currents

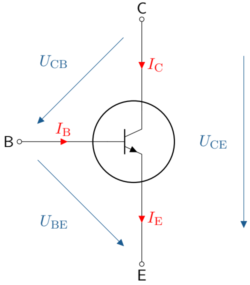


- Values in the switched-on state:
- $U_{\mathrm{BE}} \approx 0.6\,\mathrm{V}$ (Base-emitter voltage)
- $U_{\mathrm{CB}} \approx 2...300\,\mathrm{V}$ (Collector base voltage)
- $I_{\mathrm{B}} = 1\,\%; I_{\mathrm{C}} = 99\,\%; I_{\mathrm{E}} = 100\,\%$
- The node and mesh rules apply in the transistor:
- $U_{\mathrm{CE}} = U_{\mathrm{CB}} + U_{\mathrm{BE}} = 0$
- $I_{\mathrm{B}} + I_{\mathrm{C}} - I_{\mathrm{E}} = 0$



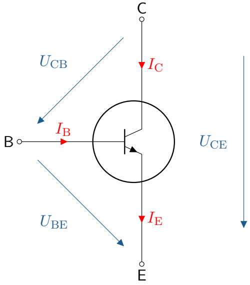
- Values in the switched-on state:
- $U_{BE} \approx 0.6\,V$ (Base-emitter voltage)
- $U_{CB} \approx 2...300\,V$ (Collector base voltage)
- $I_B = 1\,\%; I_C = 99\,\%; I_E = 100\,\%$
- The node and mesh rules apply in the transistor:
- $U_{CE} = U_{CB} + U_{BE} = 0$
- $I_B + I_C - I_E = 0$
- Transistor amplifies current and voltage:

Transistor voltages and currents



- Values in the switched-on state:
- $U_{\mathrm{BE}} \approx 0.6\,\mathrm{V}$ (Base-emitter voltage)
- $U_{\mathrm{CB}} \approx 2...300\,\mathrm{V}$ (Collector base voltage)
- $I_{\mathrm{B}} = 1\,\%; I_{\mathrm{C}} = 99\,\%; I_{\mathrm{E}} = 100\,\%$
- The node and mesh rules apply in the transistor:
- $U_{\mathrm{CE}} = U_{\mathrm{CB}} + U_{\mathrm{BE}} = 0$
- $I_{\mathrm{B}} + I_{\mathrm{C}} - I_{\mathrm{E}} = 0$
- Transistor amplifies current and voltage:
- $B = \frac{I_{\mathrm{C}}}{I_{\mathrm{B}}}$

Transistor voltages and currents



- Values in the switched-on state:
- $U_{BE} \approx 0.6\,\mathrm{V}$ (Base-emitter voltage)
- $U_{CB} \approx 2...300\,\mathrm{V}$ (Collector base voltage)
- $I_B = 1\,\%; I_C = 99\,\%; I_E = 100\,\%$
- The node and mesh rules apply in the transistor:
- $U_{CE} = U_{CB} + U_{BE} = 0$
- $I_B + I_C - I_E = 0$
- Transistor amplifies current and voltage:
- $B = \frac{I_C}{I_B}$
- $D = \frac{\Delta U_{BE}}{\Delta U_{CE}}$

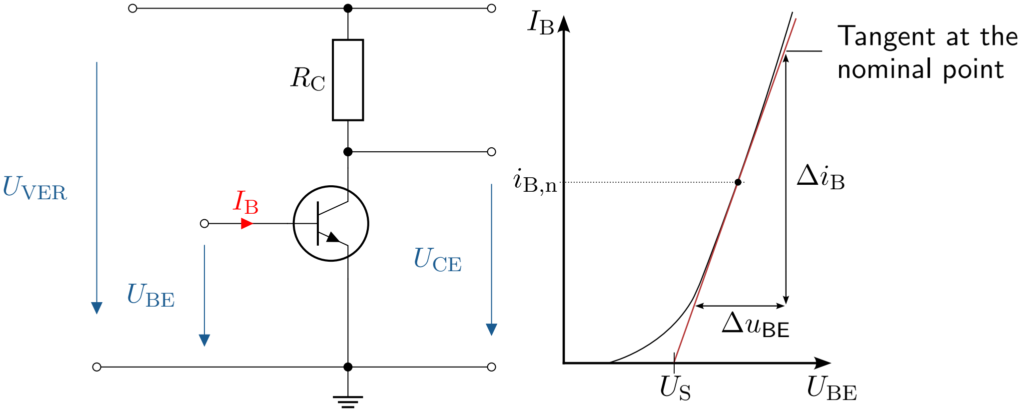


Figure: **Input characteristic curve of an NPN transistor.**

- Relationship between base-emitter voltage U_{BE} and base current I_B at constant output voltage U_{CE}.

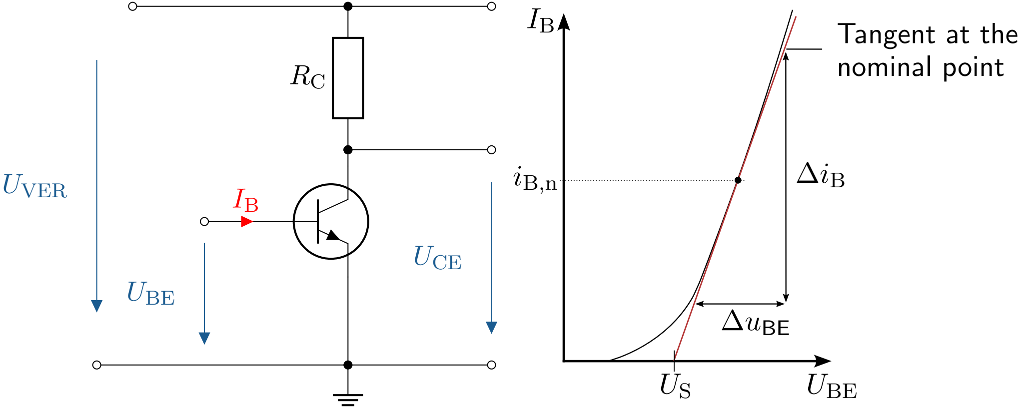


Figure: **Input characteristic curve of an NPN transistor.**

- Relationship between base-emitter voltage U_{BE} and base current I_B at constant output voltage U_{CE}.
- At the operating point, the differential resistance is defined as $r_{BE} = \frac{\Delta U_{BE}}{\Delta I_B}$.

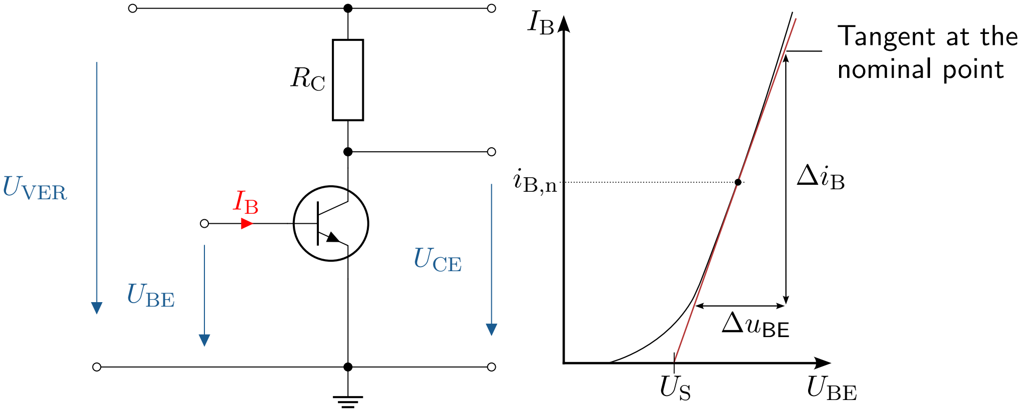


Figure: **Input characteristic curve of an NPN transistor.**

- Relationship between base-emitter voltage U_{BE} and base current I_B at constant output voltage U_{CE}.
- At the operating point, the differential resistance is defined as $r_{BE} = \frac{\Delta U_{BE}}{\Delta I_B}$.
- Compare diode characteristic curve

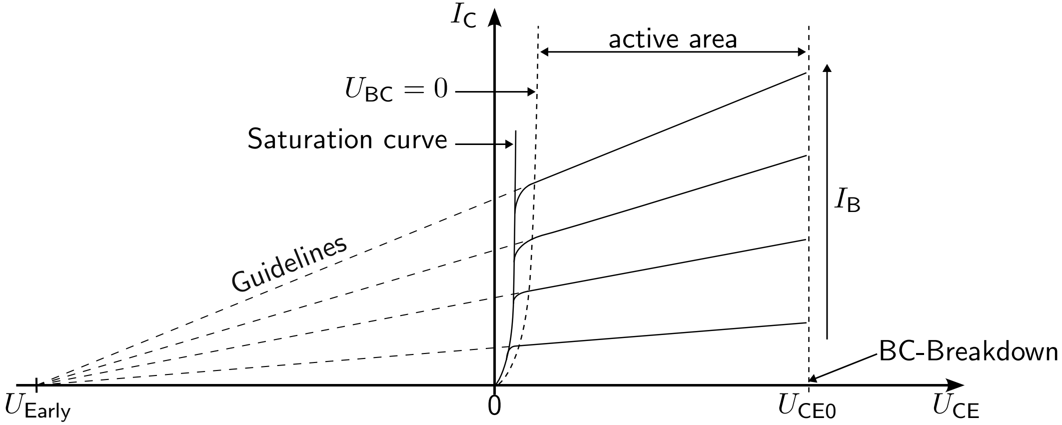
Key point:

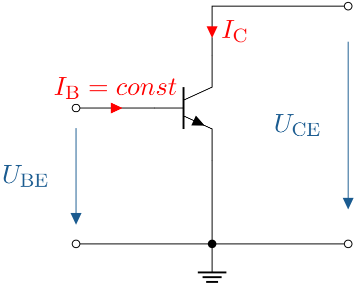
- The input characteristic curve $I_\mathrm{B} = f(U_\mathrm{BE})$ is similar to that of a diode.

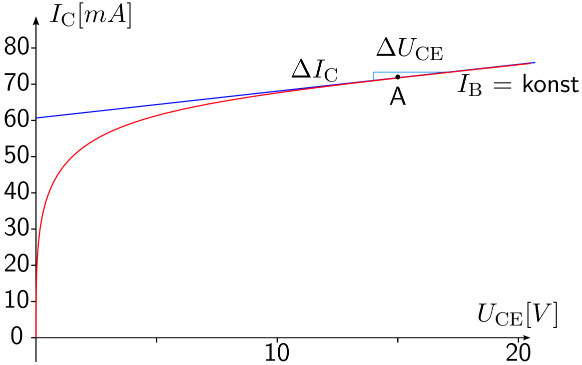
Key point:

- The input characteristic curve $I_{\mathrm{B}} = f(U_{\mathrm{BE}})$ is similar to that of a diode.
- A change in U_{CE} leads to a shift in the characteristic curve.

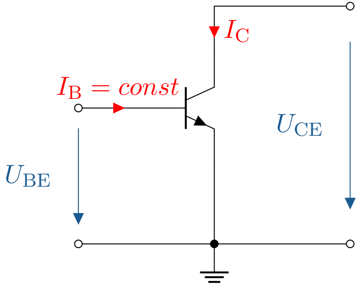
Bipolar transistor – Electrical behavior

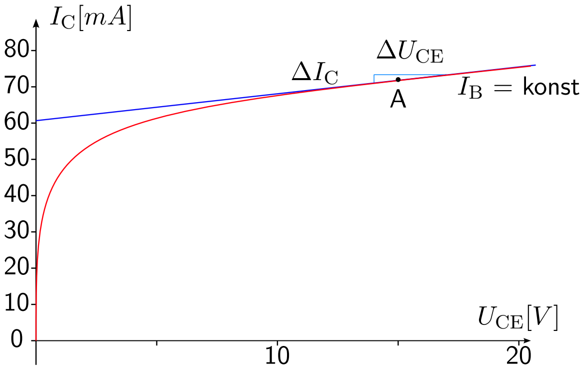






- Relationship between collector-emitter voltage U_{CE} and collector current I_C at constant base current I_B.





- Relationship between collector-emitter voltage U_{CE} and collector current I_C at constant base current I_B.
- Differential resistance $r_{CE} = \frac{\Delta U_{CE}}{\Delta I_C}$.

Key point:

- The output characteristic curve represents $I_{\mathrm{C}} = f(U_{\mathrm{CE}})$.

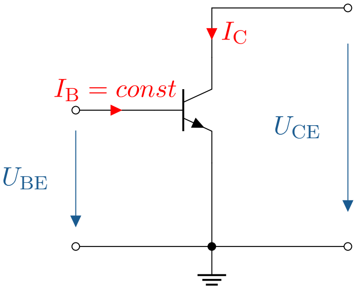
Key point:

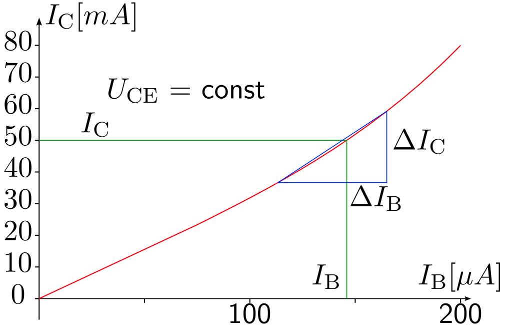
- The output characteristic curve represents $I_{\mathrm{C}} = f(U_{\mathrm{CE}})$.
- In the active range, the curve flattens out and the influence of U_{CE} on I_{C} decreases.

Key point:

- The output characteristic curve represents $I_{\mathrm{C}} = f(U_{\mathrm{CE}})$.
- In the active range, the curve flattens out and the influence of U_{CE} on I_{C} decreases.
- I_{B} affects the height of the characteristic curve.

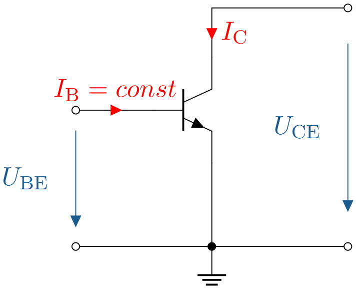
Current control characteristic

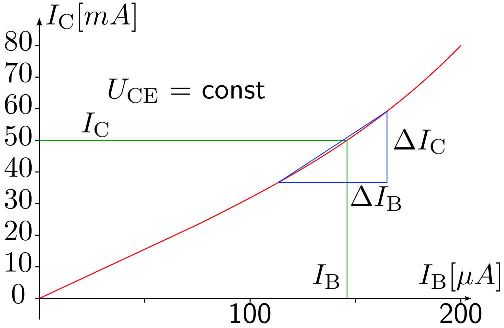




- Relationship between base current I_B and collector current I_C at constant collector-emitter voltage U_{CE}.

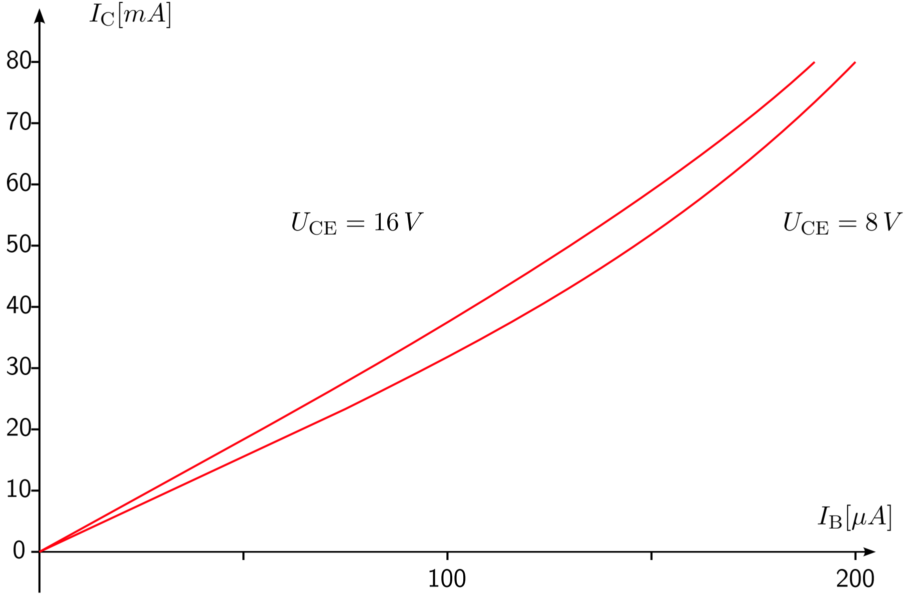
Current control characteristic

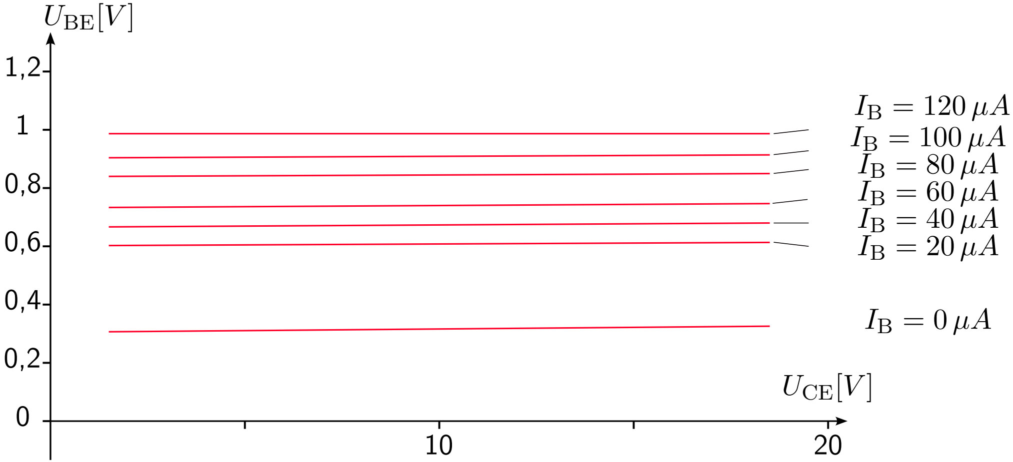


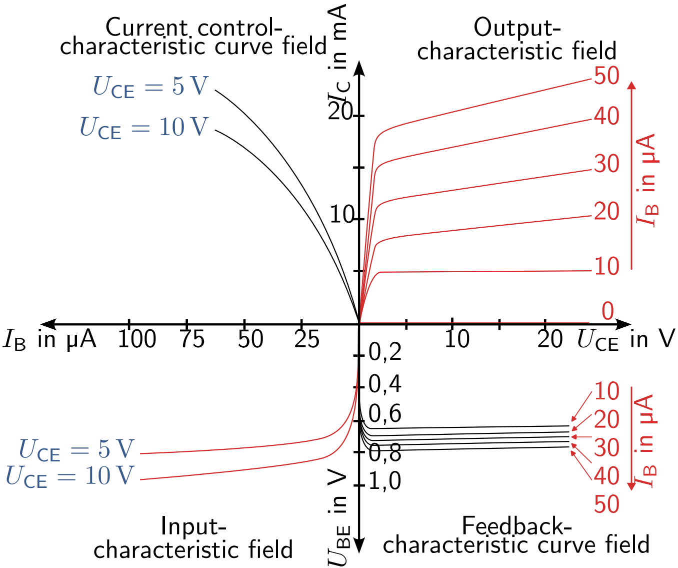


- Relationship between base current I_B and collector current I_C at constant collector-emitter voltage U_{CE}.
- Not a straight line: initially linear increase, then the slope increases with increasing base current.

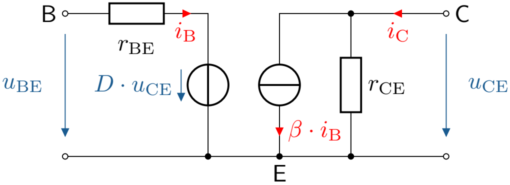
- Set of curves for variable collector-emitter voltage U_{CE}.

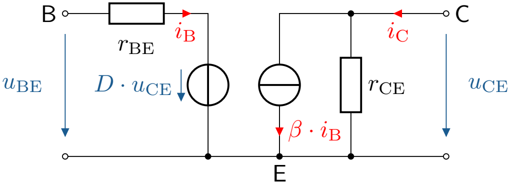


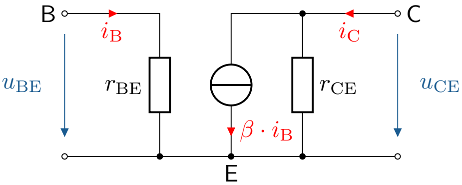

$U_{\mathrm{BE}}[V]$
1,2
1
0,8
0,6
0,4
0,2
0
10
20
$U_{\mathrm{CE}}[V]$
$I_{\mathrm{B}} = 120\,\mu A$
$I_{\mathrm{B}} = 100\,\mu A$
$I_{\mathrm{B}} = 80\,\mu A$
$I_{\mathrm{B}} = 60\,\mu A$
$I_{\mathrm{B}} = 40\,\mu A$
$I_{\mathrm{B}} = 20\,\mu A$
$I_{\mathrm{B}} = 0\,\mu A$



Current control-
characteristic curve field
$U_{CE} = 5\,V$
$U_{CE} = 10\,V$
I_C in mA
20
10
Output-
characteristic field
50
40
30
20
10
0
I_B in μA
I_B in μA
100
75
50
25
10
20
U_{CE} in V
0,2
0,4
0,6
0,8
1,0
$U_{CE} = 5\,V$
$U_{CE} = 10\,V$
10
20
30
40
50
I_B in μA
U_{BE} in V
Input-
characteristic field
Feedback-
characteristic curve field


B
C
E
r_{BE}
i_{B}
i_{C}
u_{BE}
$D \cdot u_{\mathrm{CE}}$
r_{CE}
u_{CE}
$\beta \cdot i_{\mathrm{B}}$


B
C
E
r_{BE}
i_B
i_C
u_{BE}
$D \cdot u_{CE}$
r_{CE}
u_{CE}
$\beta \cdot i_B$


B
C
E
i_B
i_C
u_{BE}
r_{BE}
r_{CE}
u_{CE}
$\beta \cdot i_B$

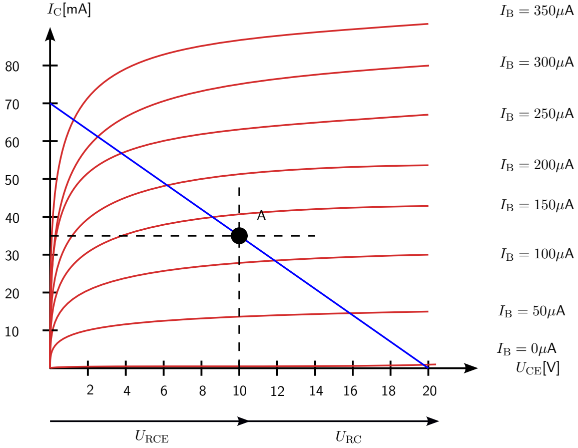
Key point:

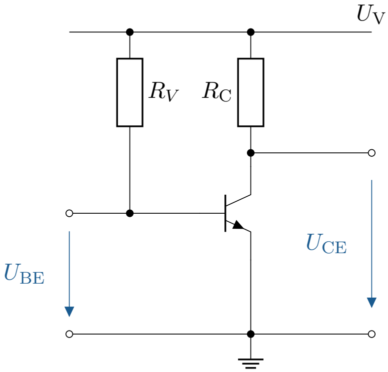
- The four-quadrant characteristic curve field shows the current-voltage characteristic curves at the input and output as well as the couplings between them.

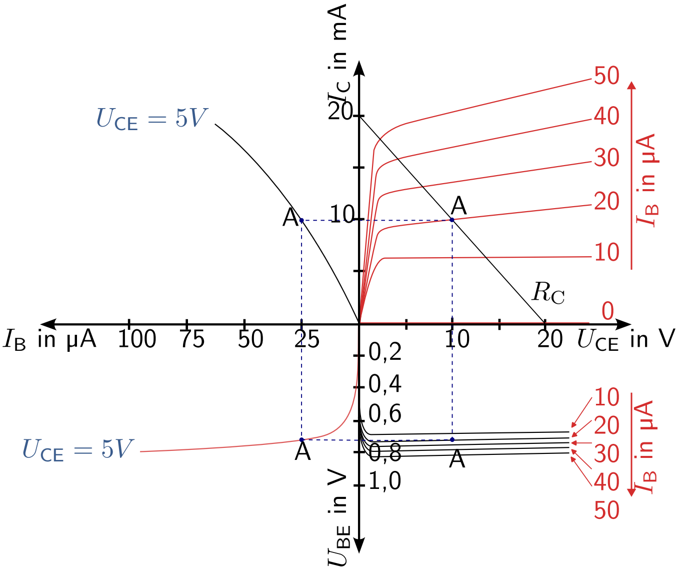
Key point:

- The four-quadrant characteristic curve field shows the current-voltage characteristic curves at the input and output as well as the couplings between them.
- The equivalent circuit diagram can be drawn up using the differential quantities.

The supply voltage U_V is $20\,\mathrm{V}$. Half of the supply voltage should be applied across the resistor R_C, which represents an ohmic load, and a current of $10\,\mathrm{mA}$ should flow.







I_C in mA
20
10
$U_{CE} = 5V$
A
A
50
40
30
20
10
0
I_B in µA
R_C
I_B in µA
100
75
50
25
10
20
U_{CE} in V
0,2
0,4
0,6
0,8
1,0
U_{BE} in V
$U_{CE} = 5V$
A
A
10
20
30
40
50
I_B in µA

Bipolar transistor – Example 2.2

The following operating points and parameters result in the respective quadrants.

Output (bei: $I_{\mathrm{B}} = 20\,\mu\mathrm{A}$):

$$
\begin{aligned}
U_{\mathrm{CE}} &= 10\,\mathrm{V} \\
I_{\mathrm{C}} &= 10\,\mathrm{mA} \\
R_{\mathrm{C}} &= \frac{U_{\mathrm{RC}}}{I_{\mathrm{RC}}} = \frac{U_{\mathrm{V}}/2}{I_{\mathrm{C}}} = \frac{10\,\mathrm{V}}{10\,\mathrm{mA}} = 1000\,\Omega
\end{aligned}
$$

The following operating points and parameters result in the respective quadrants.

Output (bei: $I_\mathrm{B} = 20\,\mu\mathrm{A}$):

$$U_\mathrm{CE} = 10\,\mathrm{V}$$
$$I_\mathrm{C} = 10\,\mathrm{mA}$$
$$R_\mathrm{C} = \frac{U_\mathrm{RC}}{I_\mathrm{RC}} = \frac{U_\mathrm{V}/2}{I_\mathrm{C}} = \frac{10\,\mathrm{V}}{10\,\mathrm{mA}} = 1000\,\Omega$$

Current control:

$$I_\mathrm{C} = 10\,\mathrm{mA}$$
$$I_\mathrm{B} = 25\,\mu\mathrm{A}$$
$$B = \frac{I_\mathrm{C}}{I_\mathrm{B}} = \frac{10\,\mathrm{mA}}{25\,\mu\mathrm{A}} = 400$$

Bipolar transistor – Example 2.2

In den jeweiligen Quadranten ergeben sich die folgenden Arbeitspunkte und Kenngrößen.

Input:

$$
\begin{aligned}
I_{\mathrm{B}} &= 25\,\mu\mathrm{A} \\
U_{\mathrm{BE}} &= 0{,}72\,\mathrm{V} \\
R_{\mathrm{V}} &= \frac{U_{\mathrm{RV}}}{I_{\mathrm{RV}}} = \frac{U_{\mathrm{V}} - U_{\mathrm{BE}}}{I_{\mathrm{B}}} = \frac{20\,\mathrm{V} - 0{,}72\,\mathrm{V}}{25\,\mu\mathrm{A}} = 771{,}2\,\mathrm{k}\Omega
\end{aligned}
$$

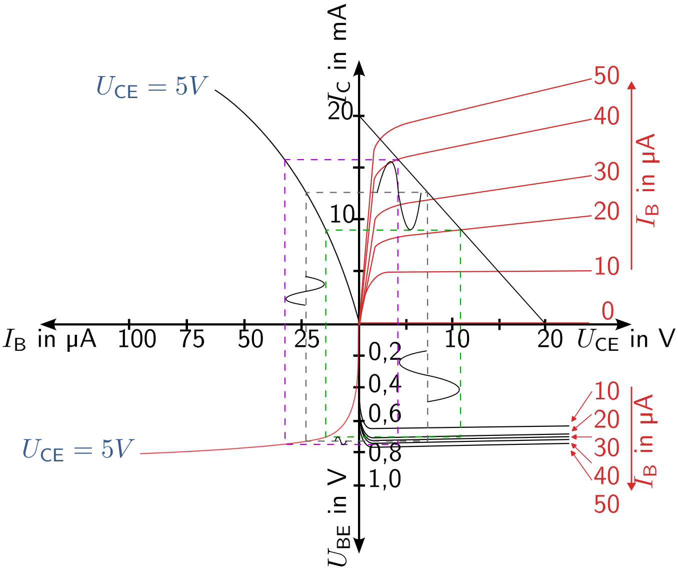
In den jeweiligen Quadranten ergeben sich die folgenden Arbeitspunkte und Kenngrößen.

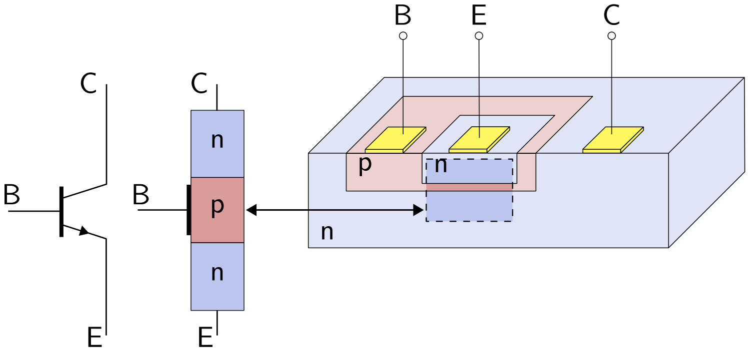
Input:

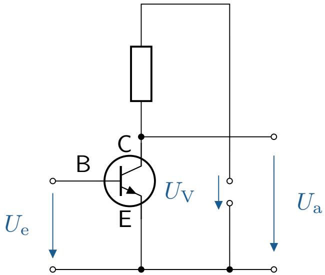
$$I_{\mathrm{B}} = 25\,\mu\mathrm{A}$$
$$U_{\mathrm{BE}} = 0{,}72\,\mathrm{V}$$
$$R_{\mathrm{V}} = \frac{U_{\mathrm{RV}}}{I_{\mathrm{RV}}} = \frac{U_{\mathrm{V}} - U_{\mathrm{BE}}}{I_{\mathrm{B}}} = \frac{20\,\mathrm{V} - 0{,}72\,\mathrm{V}}{25\,\mu\mathrm{A}} = 771{,}2\,\mathrm{k\Omega}$$

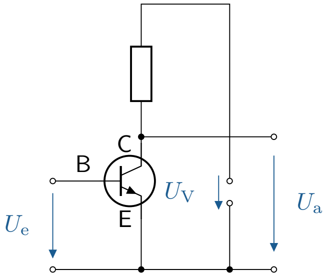
Retroactive effect:

$$U_{\mathrm{BE}} = 0{,}72\,\mathrm{V}$$
$$U_{\mathrm{CE}} = 10\,\mathrm{V}$$
$$D = \frac{U_{\mathrm{BE}}}{U_{\mathrm{CE}}} = \frac{0{,}72\,\mathrm{V}}{10\,\mathrm{V}} = 0{,}072$$

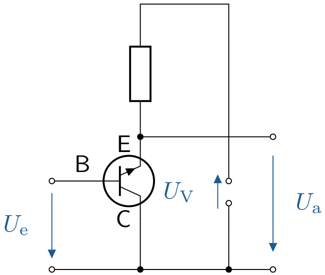

$U_{CE} = 5V$
I_C in mA
20
10
50
40
30
20
10
0
I_B in µA
I_B in µA
100
75
50
25
10
20
U_{CE} in V
0,2
0,4
0,6
0,8
1,0
U_{BE} in V
10
20
30
40
50
I_B in µA
$U_{CE} = 5V$


C
B
E
C
n
B
p
n
E
B
E
C
p
n
n


B
C
E
U_V
U_e
U_a



C
B
E
U_V
U_a
U_e



E
B
C
U_V
U_a
U_e

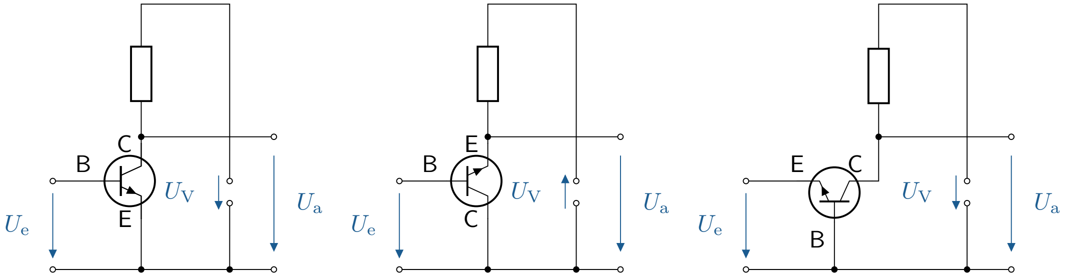


Figure: **Basic circuits npn bipolar transistor.** From left to right: emitter, collector, and base circuit.

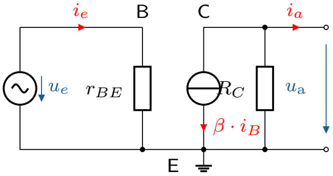


Figure: **Replacement circuit diagrams for the basic circuits.** From left to right: emitter, collector, and base circuits of npn transistors.

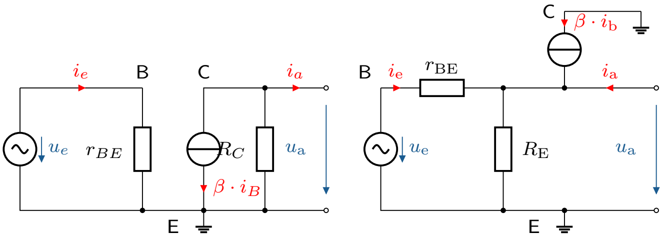


Figure: **Replacement circuit diagrams for the basic circuits.** From left to right: emitter, collector, and base circuits of npn transistors.

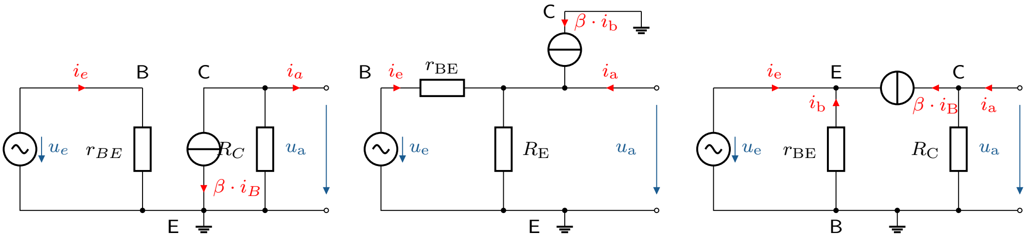


Figure: **Replacement circuit diagrams for the basic circuits.** From left to right: emitter, collector, and base circuits of npn transistors.

r_e	Average e.g. $1\mathrm{k}\Omega$	Small e.g. 50Ω	large e.g. $100\mathrm{k}\Omega$
r_a	Average e.g. $10\mathrm{k}\Omega$	large e.g. $100\mathrm{k}\Omega$	Small e.g. 50Ω
v_i	large e.g. 100	<1 e.g. 0.9	large e.g. 100
v_u	large e.g. 100	large e.g. 100	<1 e.g. 0.99
v_p	very large e.g. $1\mathrm{k}\Omega$	large e.g. 100	large e.g. 100
φ_u	antiphase 180°	in phase 0°	in phase 0°

Tabelle: Electrical properties of basic circuits

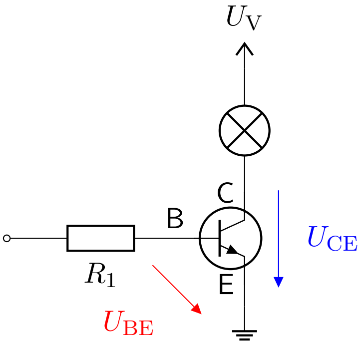
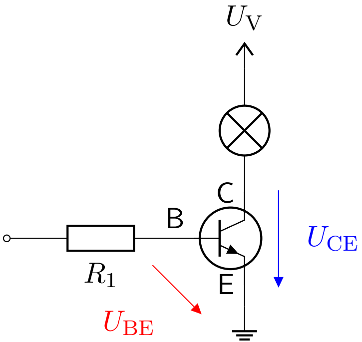


Figure: **Transistor as a switch.** Left: Circuit diagram with lamp as load in the collector-emitter path. Right: Voltage curves with U_{CE} as a function of U_{BE}.



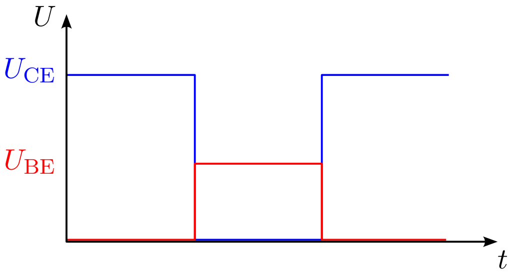
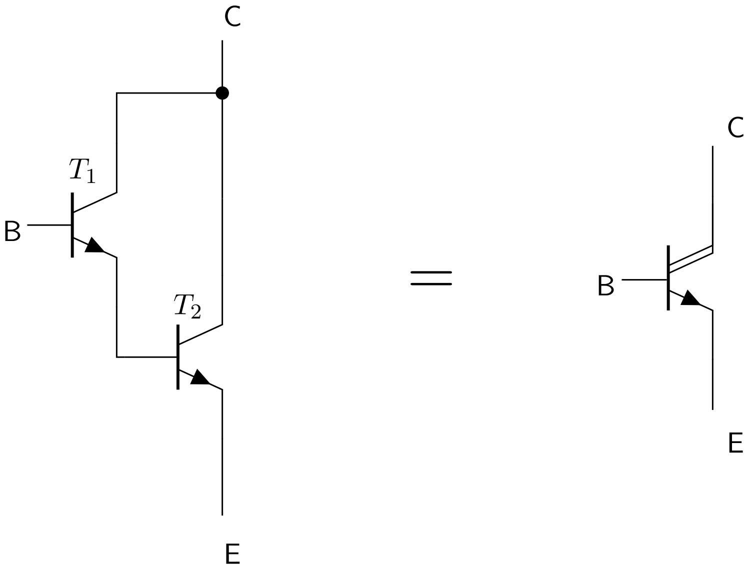
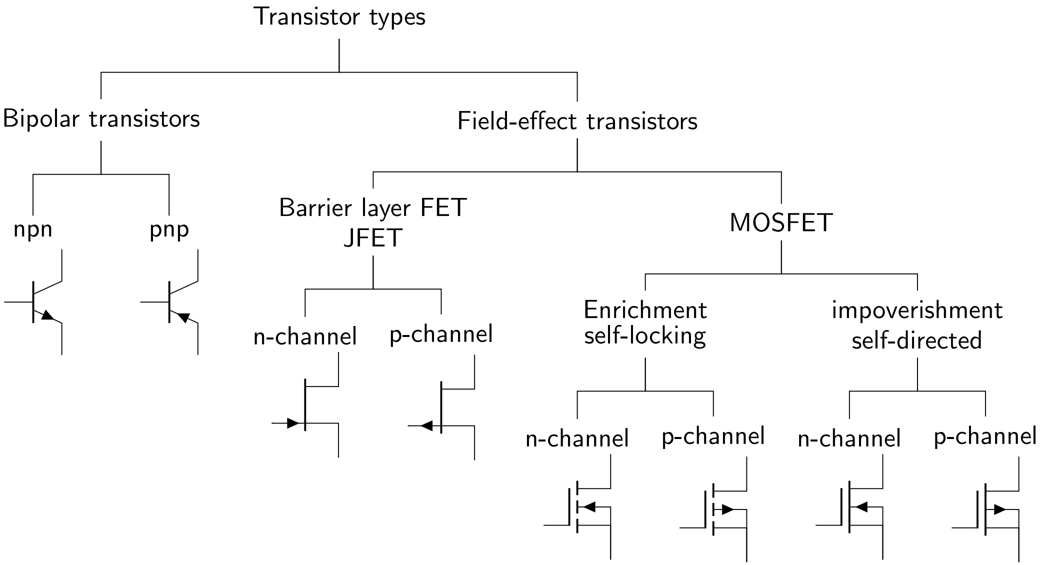


Figure: **Transistor as a switch.** Left: Circuit diagram with lamp as load in the collector-emitter path. Right: Voltage curves with U_{CE} as a function of U_{BE}.

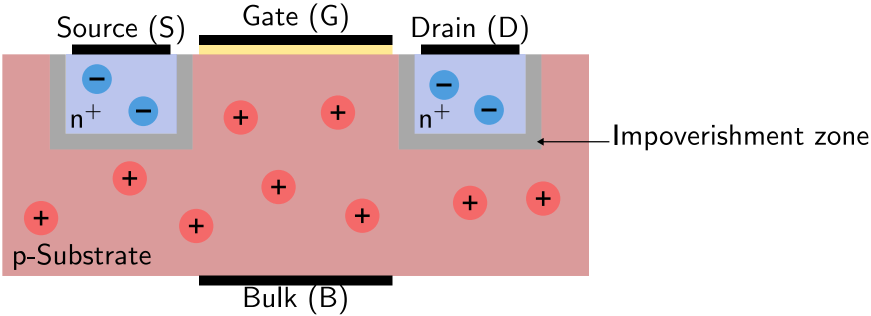
Key point:
Compared to an electromechanical switch in the form of a relay, transistors have a significantly smaller installation space and lower price. Due to the contactless design, there is also no contact bounce, which leads to a longer service life.

Bipolar transistor – Application/Basic circuits

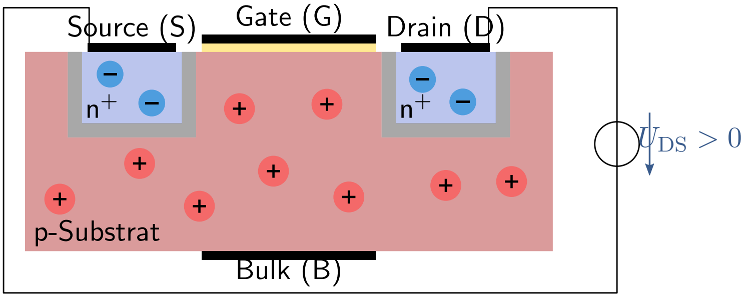


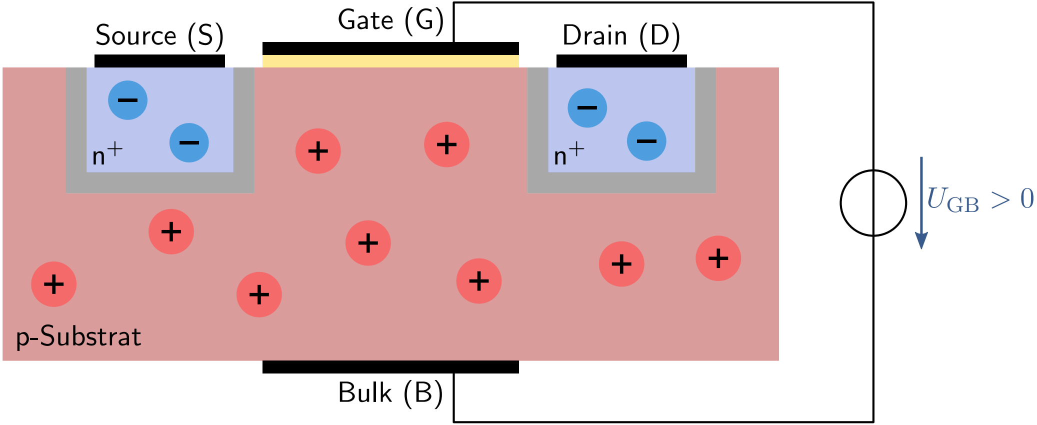

Transistor types
Bipolar transistors
Field-effect transistors
npn
pnp
Barrier layer FET
JFET
MOSFET
n-channel
p-channel
Enrichment
self-locking
impoverishment
self-directed
n-channel
p-channel
n-channel
p-channel

Field effect transistor – Electrical behavior



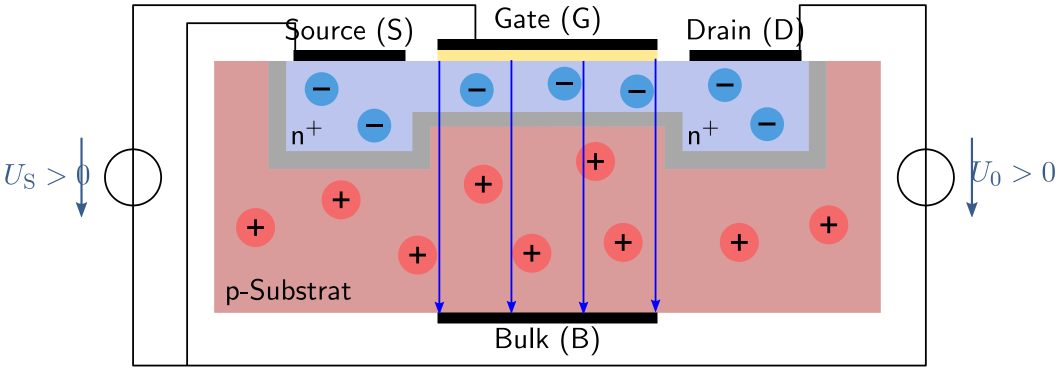
Field effect transistor – Electrical behavior

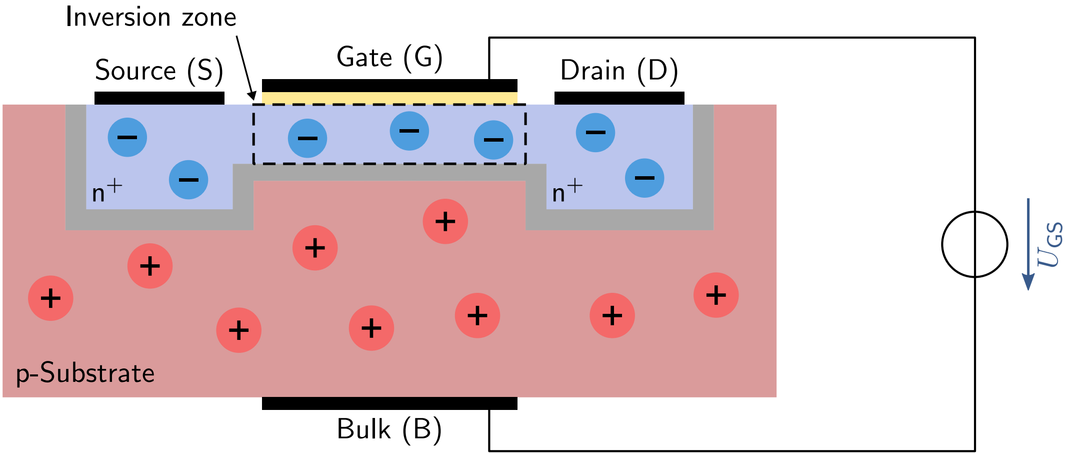




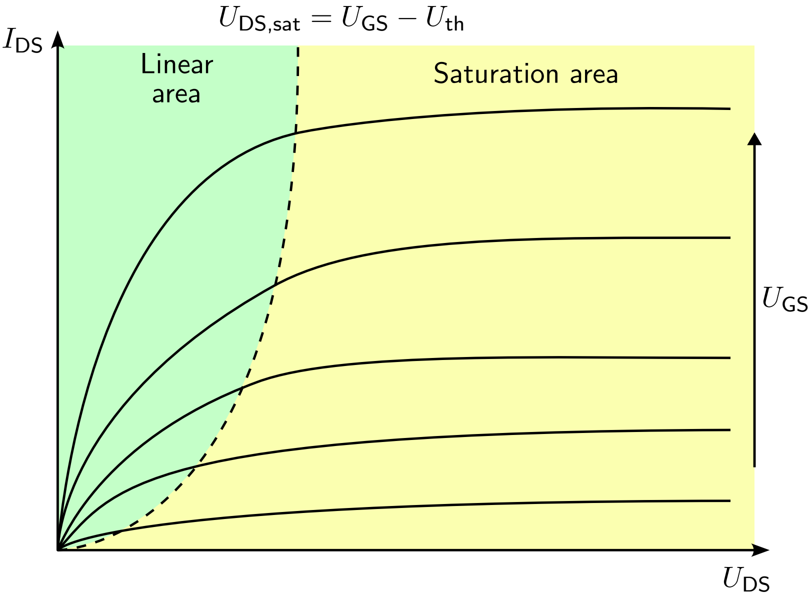
Gate (G)
Source (S)
Drain (D)
n+
n+
$U_{\mathrm{GB}} > 0$
p-Substrat
Bulk (B)

Field effect transistor – Electrical behavior

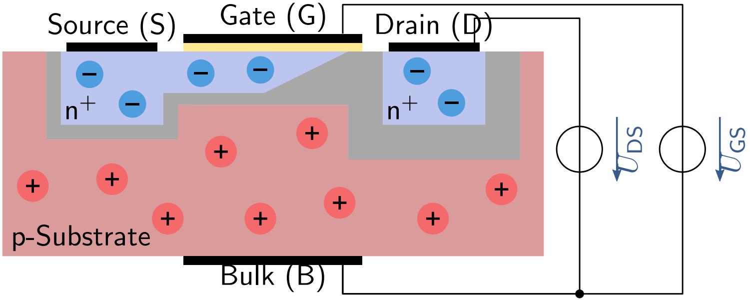




Inversion zone
Source (S)
Gate (G)
Drain (D)
n+
n+
p-Substrate
Bulk (B)
U_{GS}

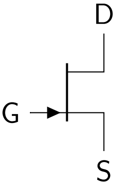
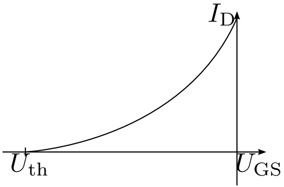
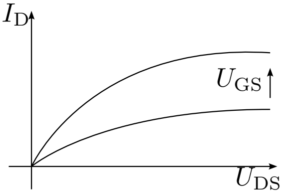
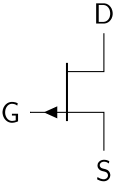
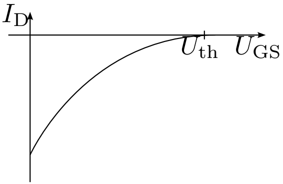
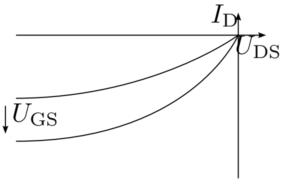


$U_{\text{DS,sat}} = U_{\text{GS}} - U_{\text{th}}$
I_{DS}
Linear area
Saturation area
U_{GS}
U_{DS}


Source (S)
Gate (G)
Drain (D)
n^+
n^+
p-Substrate
Bulk (B)
U_{DS}
U_{GS}

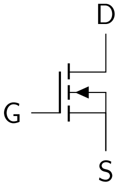
Key point:

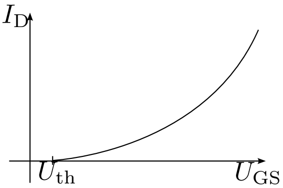
- MOSFETs have three terminals: gate (G), source (S), and drain (D).
- The electrical behavior is controlled without power.
- U_{GS} causes an inversion of the charge carriers below the gate.

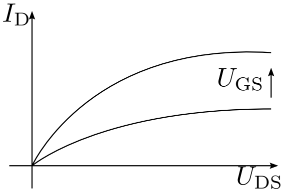
Type	Symbol	Input characteristic curve	Output characteristic field
n-JFET			
p-JFET			

Field effect transistor – Electrical behavior

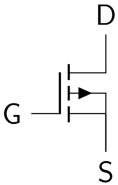
n-Mosfet, self-locking

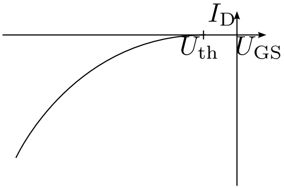


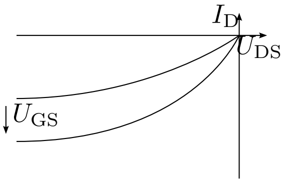




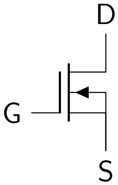
p-Mosfet, self-locking

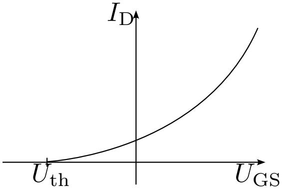


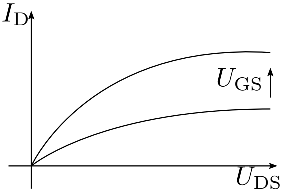




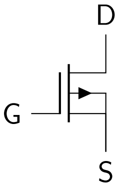
n-Mosfet, self-directed

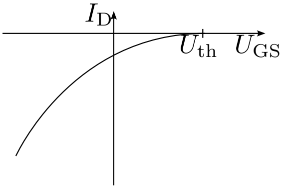


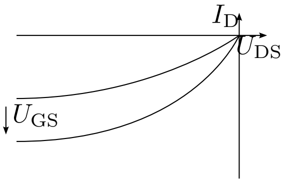




p-Mosfet, self-directed



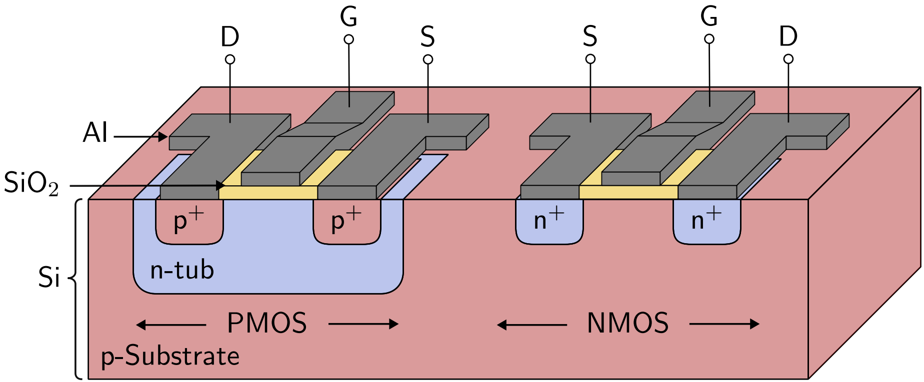




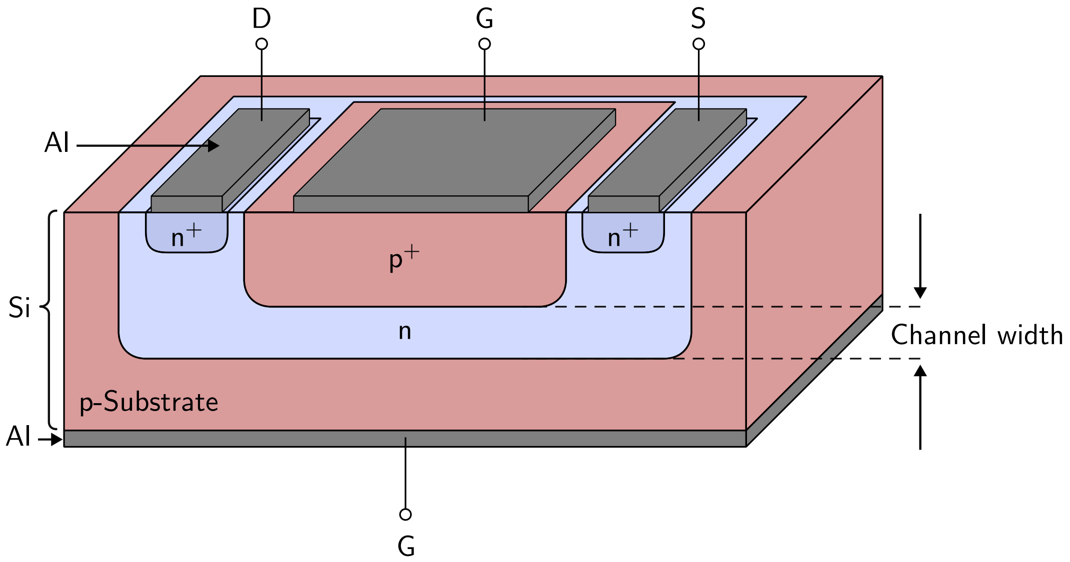
Key point:

- The channel bias determines the necessary signs of the currents and voltages at the transistor.
- The input characteristics can be distinguished by the position of the threshold voltage U_{th}.

Field effect transistor – Structure

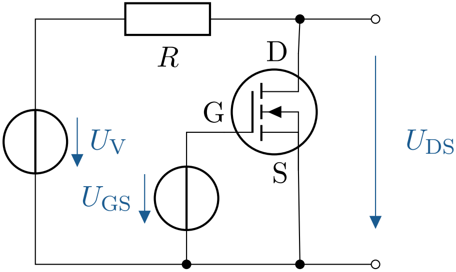


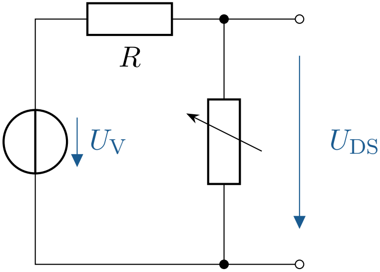
Field effect transistor – Structure



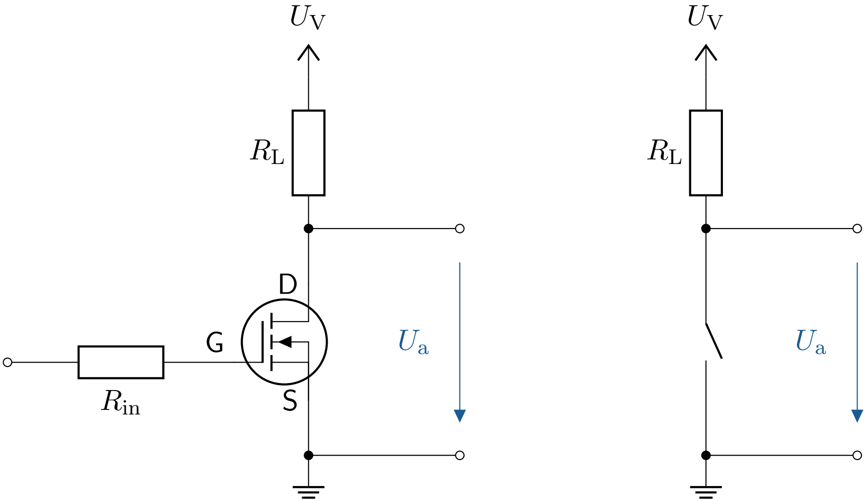
Key point:

- CMOS technology combines complementary MOSFETs in a single component.
- JFETs have the simplest FET structure.

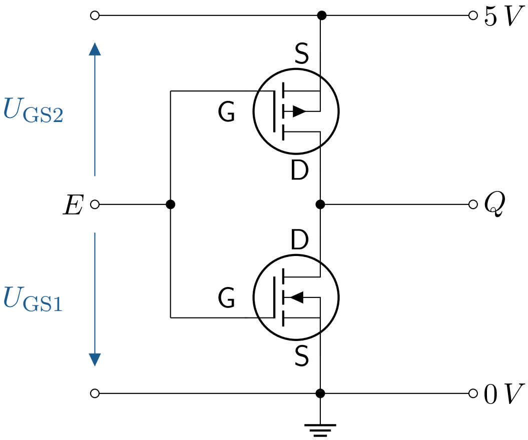

R
D
G
S
U_V
U_{GS}
U_{DS}

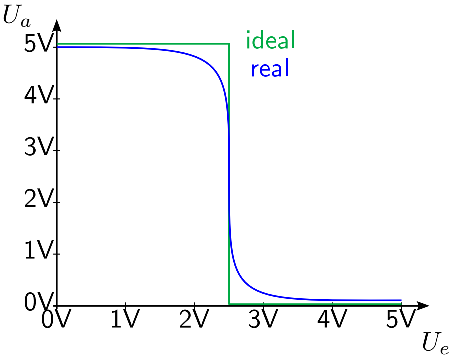

R
U_V
U_{DS}

Key point: The ohmic resistance of a MOSFET can be controlled by U_{GS}, but the conduction resistance is low and the value range is very narrow.


U_V
R_L
D
G
S
R_{in}
U_a
U_V
R_L
U_a

Key point: The major advantage of MOSFETs in this scenario is their high switching speed, low resistance in the conductive state, and high input resistance at the gate.


5 V
S
U_GS2
G
D
E
Q
D
U_GS1
G
S
0 V


U_a
ideal
real
5V
4V
3V
2V
1V
0V
0V
1V
2V
3V
4V
5V
U_e